

INSPECTION & SPECIFICATION SERVICES CC.

CONTROLLED AUDIT INSPECTION REPORT

Combined-Product Spillage Inspection Assessment Report
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Combined-Product Spillage Inspection Assessment Report

South32 Wessels Mine · Surface and Underground Conveyor / Process Areas

Audit report · engineering assessment · maintenance recommendations



Client	South32 – Wessels Mine	Mine / site	Wessels Mine
Area / system	Combined-product handling and processing systems - selected surface and underground sections	Revision	1.5
Report number	S32WPPSR-01	Inspection date	July 2026
Issue date	01/09/2026	Prepared by	Brendan Wood
Issue status	Approved	Approved by	Kevin Wood
Reviewed by	Kevin Wood	ISS job number	IRS32-Wessels-0726_1
Client reference	South32 Wessels		

CONFIDENTIALITY & RELIANCE

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Document Control and Metadata

Table 1 — Document control and report metadata

Field	Value
Report title	Combined-Product Spillage Inspection and Engineering Assessment Report
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Prepared, Reviewed and Approved

Table 2 — Document authorization record

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Revision History

Table 3 — Revision history

Revision	Date	Change description	Prepared by	Approved by
0.1	15 Aug 2026	Initial spillage-report working base adapted from ISS sheave-report structure using consolidated Contextual Notes V3.7 and completed evidence review.	ISS	Draft review
0.2	25 Aug 2026	Phase 6 assembly draft: accepted narrative architecture + evidence hierarchy + CV09 holistic evidence grammar integrated into one full report structure.	ISS	Draft review
0.3	25 Aug 2026	Phase 7 compilation draft: refined Parts I-XII, mature action programme, verification register, technical appendices and current conceptual engineering figures compiled into the approved V0.2 report architecture.	ISS	Draft review
0.4	25 Aug 2026	Language-harmonized revision: V0.3 engineering substance retained while reader-facing narrative was aligned to the accepted V0.2 softer formal-engineering style under Context V3.12.	ISS	Draft review

Revision	Date	Change description	Prepared by	Approved by
0.5	25 Aug 2026	Global terminology refinement: reader-facing uses of “envelope” were replaced with direct physical terms such as material-handling path, carrying path, process stream/process path and installation space. “Envelope” is retained only where it describes a defined geometric or operating range, such as belt tracking or normal operating envelope.	ISS	Draft review
0.6	28 Aug 2026	Stage-B integrated controlled revision: V0.5 reconciled against source-reconciled Context V4.3, Stage-A gap register, current evidence overlays and locked functional engineering / operational-control concepts. Superseded evidence and conceptual interpretations removed from active report use.	ISS	Draft review
0.6.1	31 Aug 2026	System-level fines-containment alignment revision: records fines spillage throughout the inspected conveyor ecosystem; establishes CV09 as the pilot for a common cleaner-and-containment functional standard; strengthens systematic review/progressive application at relevant head/discharge locations; records the CV09 downstream fines-handling endpoint; aligns Priority 4 and the conclusion; retains location-specific mechanism and geometry qualifications.	ISS	Draft review
1.5	1 September 2026	Final controlled issue revision: incorporates final engineering-content reconciliation, evidence/source-control closure, body video hyperlinks and QR access, cover/document-control completion and technical review/issue sign-off	ISS	Kevin Wood

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PART I — WHY THE AUDIT WAS UNDERTAKEN

1 Executive Summary

Fines spillage was observed throughout the inspected Wessels material-handling system, from the underground conveyor route through the surface screening and loading-station areas, although the extent varied considerably between locations. No inspected conveyor section was completely free of fines accumulation. The audit was undertaken to determine whether these deposits were isolated housekeeping problems or evidence of broader shortcomings in retaining combined product and process water within their intended handling and process paths.

The evidence does not support one universal spillage mechanism. The presence of fines was system-wide, while the severity, local condition and immediate mechanism were not necessarily the same at every location. Carryback remains the strongest recurring mechanism identified across the system, but the report does not assume that every individual fines deposit arose from the same sole cause. Other important local mechanisms include coarse or mixed product leaving the usable carrying area, belt tracking/alignment concerns, feed/loading and product-size constraints associated with the limited carrying width of the Sample Conveyors, active process-water escape at the Screens, structural deterioration, and intermittent event-driven product removal associated with trips or restart preparation.

A central finding is that the location where material accumulates is not necessarily the location where the initiating failure occurs. Residual fines can remain on the return belt after discharge and detach progressively downstream; rollers/idlers can dislodge additional adhered material; water can mobilize fines away from an original escape point; mistracking can reduce carrying margin before product is lost; and trip/restart clearing can create a temporary deposit that is operationally different from chronic spillage.

Taken together, the evidence indicates that increased housekeeping alone will not provide a sustainable long-term solution. Housekeeping remains necessary for access, recovery and consequence control, but recurring deposits require the underlying source mechanism or primary contributor to be addressed where it can be established.

AUDIT TAKEAWAY

The inspected Wessels material-handling system exhibits a material-containment problem involving several mechanisms, rather than simply a housekeeping problem or a collection of unrelated scraper defects.

The higher-order engineering objective is therefore to keep combined product — fines and coarse/lump ore — within the intended material-handling path throughout transit and to retain process water within the screening and processing systems where it is intended to remain. The physical control should match the actual mechanism and local geometry rather than impose one universal hardware solution.

The recommended programme combines system-level source control with targeted local actions. CV09 remains the preferred pilot for demonstrating a pulley-supported cleaner-and-containment function inside the head/discharge chute so that residual fines are removed before the return run and cleaner-removed material remains in the existing discharge flow. Once that function has been demonstrated under representative operation, the relevant head/discharge locations across the inspected conveyor route should be systematically reviewed against the same functional standard and progressively adapted where the local geometry, operating duty and maintenance requirements confirm applicability.

W10 requires belt-edge side skirting/spill sealing as the preferred engineered containment addition for the demonstrated lateral coarse/mixed-product loss within the restricted tunnel environment. Representative loading, belt tracking and local conveyor geometry should still be verified and confirmed contributors corrected so that the final skirting arrangement is developed around the verified operating condition. The same side-skirt/spill-seal principle is proposed for Sample Conveyors 2 and 3 after their feed quantity/variation, product-size distribution, loading condition and belt-position/tracking or alignment have been quantified and corrected or adjusted where required. If the W10 principle performs successfully, it should also be considered and adapted at other underground or surface conveyor locations where a comparable lateral-containment requirement is identified and the local mechanism and geometry confirm suitability.

U4 and L3 require location-specific alignment/tracking verification; the Screens require separate water-source and structural workstreams; L1 requires structural and belt-clearance assessment; and trip/restart and housekeeping controls require area-specific operational confirmation. Functional success should ultimately be demonstrated under representative operation rather than inferred from work completion or a temporarily clean area.

2 Purpose and Objectives

The purpose of the audit is to identify where combined product or process water is escaping its intended handling path, determine the mechanisms and contributing conditions associated with those losses where the available evidence allows, distinguish recurring system mechanisms from local asset-specific conditions, and define practical engineering, maintenance and operational actions that can reduce recurrence.

To achieve this purpose, the audit is structured around the following specific objectives:

- Identify the material types and locations at which visible losses occur.
- Determine the immediate mechanism and contributing condition where the available evidence supports that conclusion.
- Distinguish visually proven conditions from inspector-recorded observations, engineering interpretation and unresolved verification items.
- Develop practical source-control, containment, maintenance, alignment, structural and operational actions.
- Provide a controlled evidence record that allows engineering and maintenance personnel to trace each major conclusion back to the inspection evidence.

The audit is qualitative and mechanism-focused. It does not attempt to infer total lost tonnage or volume from photographs, and unverified numerical tolerances, feed rates, structural capacities or OEM limits are not introduced as report facts. Visible extent is used to describe the inspected condition, while values that were not measured remain items for later measurement, design development or competent assessment.

3 Scope of the Audit

The inspected locations form one connected material-handling route and are treated as an ecosystem rather than as an unrelated list of assets.

Table 5 — Audit scope — inspected process areas and included assets

Process area	Included assets / groups
Underground — Central Block	Underground Crusher; UC05 / UC04 / UC03 / UC02 / UC01 continuous conveyor system
Underground — Secondary Plant	U3 Conveyor; U4 Tripper Car
Underground — Silos / Feeders	U5A Conveyor; U5B Conveyor
Surface — Screening / Transfer	W10; CV01; CV02; CV03; CV04; CV06; CV09; CV14; CV15; Screens / screen chutes
Surface — Loading Station	L1; L2; L3; L4; Sample Conveyors 1, 2 and 3

4 Methodology, Evidence Basis and Limitations

The assessment combines site inspection observations, still photographs, inspection videos, post-inspection internal evidence-review questionnaires, the source-reconciled Context V4.3, controlled visual-review records and the current evidence-index/overlay controls. The current active inspection-media population is 196 items: 188 photographs and 8 videos, classified for report-development priority as 79 Essential body (E), 40 Supporting/story (S) and 77 Appendix/reference (A). Each item retains its canonical ID.

Table 6 — Evidence / statement classification used in this report

Evidence / statement type	Meaning in this report
Direct observation	Condition recorded directly during the inspection by the inspector.
Visual evidence-supported	Condition materially demonstrated by supplied photographs or video.
Source-recorded information	Operating, maintenance or historical information retained from inspection records/questionnaires.
Engineering interpretation / hypothesis	Reasoned explanation consistent with the observed condition but not fully proven by the available evidence.
Accepted functional remedial concept	Audit-level functional direction accepted for report development; site measurement, detailed design and implementation verification may remain open.
Unresolved / requires verification	Condition or magnitude that cannot be responsibly closed from the present evidence.
Detailed-design / competent-authority responsibility	OEM, responsible-engineer, structural authority or South32 governance detail that lies beyond the evidence-supported audit concept.

The inspection included both static observations undertaken while the relevant equipment was not operating and dynamic observations undertaken while the equipment was in operation. Access during the operating observations was necessarily more restricted in some areas because of the proximity to operating industrial equipment. Where belt geometry, offset magnitude, cleaner contact/wear travel, feed quantity, remaining structural capacity, chute dimensions, internal screen-water source or design limits were not measured, the report deliberately carries those

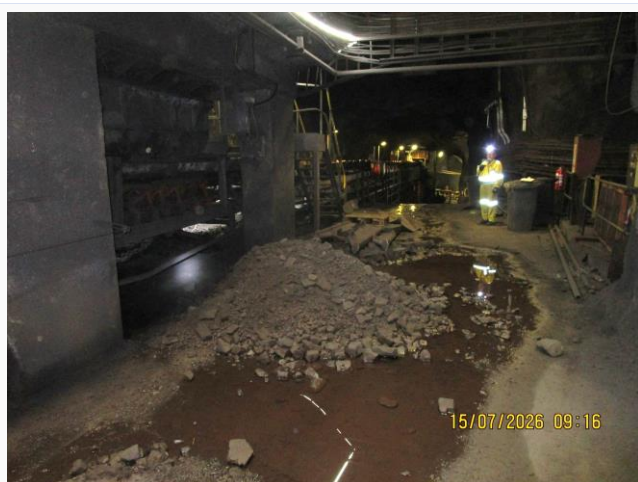
items forward for verification, detailed design or competent assessment rather than inferring a value. This applies particularly to the Sample Conveyor tracking conditions, U4 and L3 alignment, the internal Screens water-escape mechanism, Screens structural capacity and the L1 historical belt-contact/structural conditions.

PART II — UNDERSTANDING THE MATERIAL-HANDLING SYSTEM

5 The Material Journey

The material handled by the inspected system is combined product comprising lump/coarse ore together with fines. Fines are already present in the product stream and additional fines are generated through crushing and subsequent handling. Their presence is therefore not itself a defect; the audit concern begins when product or process water leaves its intended material-handling or process path before reaching the stage designed to receive or control it.

From the underground crusher onward, the conveyor system is intended to transport combined product through successive transfer and processing stages while retaining it within the designed material path. A portion of the finer material can remain adhered to the belt after discharge and travel back with the return side of the belt; this residual material is referred to as carryback. If it is not removed and retained at an appropriate discharge or cleaning point, it can progressively fall from the return belt and become distributed spillage along the conveyor route. Because fines form part of the combined product being conveyed, the objective is not simply to clean them from the belt, but to remove them in a controlled manner and keep them within the intended process path wherever practicable. Residual carryback should therefore be removed at appropriate discharge locations and kept inside the process; process water should remain within the screening/washing route; and material deliberately removed during unavoidable operating interventions should remain under controlled temporary handling and be recovered through an approved process route. Within the surface screening process, fines separated during washing are conveyed by CV09 to the downstream fines collection/handling point. This provides an important reference for the wider containment philosophy: fines should remain within the intended conveyor and process stream through successive handling stages until they reach the point at which the process is designed to separate, collect and handle them.



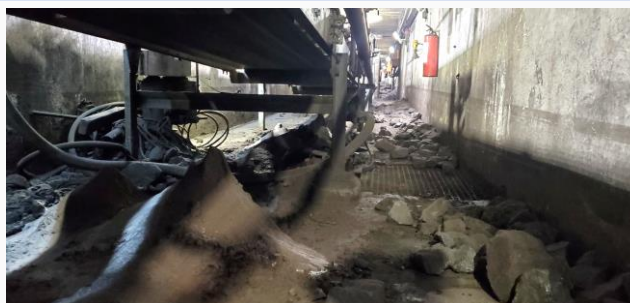
UGCrusher-REF-001 — mixed lump/fines accumulation at the underground start of the inspected conveyor ecosystem.



UGUC05_04_03_02_01-REF-006 — severe local accumulation along the continuous underground conveyor route.



CV04-REF-005 — wider surface-plant view showing multiple accumulated fines piles beneath and around the conveyor structure.



W10-REF-006 — mixed coarse product and fines distributed below the conveyor around components within the restricted tunnel area.

READER TAKEAWAY

The report concerns loss of product or process containment along the material-handling and processing journey — not the mere presence of fines, lump ore or process water within the system.

6 Central Audit Principle — Spillage Location Is Not Always Cause Location

Visible spillage should not automatically be interpreted as a local equipment failure at the point where the material is found. A deposit may be the final consequence of an initiating condition that is upstream or functionally separate from the deposit location. For this reason, the assessment considers the observed deposit, the material type, the immediate mechanism, possible contributing conditions and the most appropriate control in sequence. Where the initiating cause cannot be established from the available evidence, the report leaves that point open for further verification rather than assigning an unsupported cause.

The following examples illustrate how material or contamination can appear at a location downstream of, or otherwise separate from, the condition that initiated the loss:

1. Fines/carryback can remain on the return belt after intended discharge and detach progressively as the belt flexes over downstream rollers.
2. Process or cleaning water can mobilize fines and transport contamination away from the original escape point.
3. Belt mistracking can reduce usable carrying margin before coarse/mixed product is lost at a later location.
4. Trip/restart clearing can place product beside a conveyor even when the initiating trip, blockage or intervention occurred elsewhere.

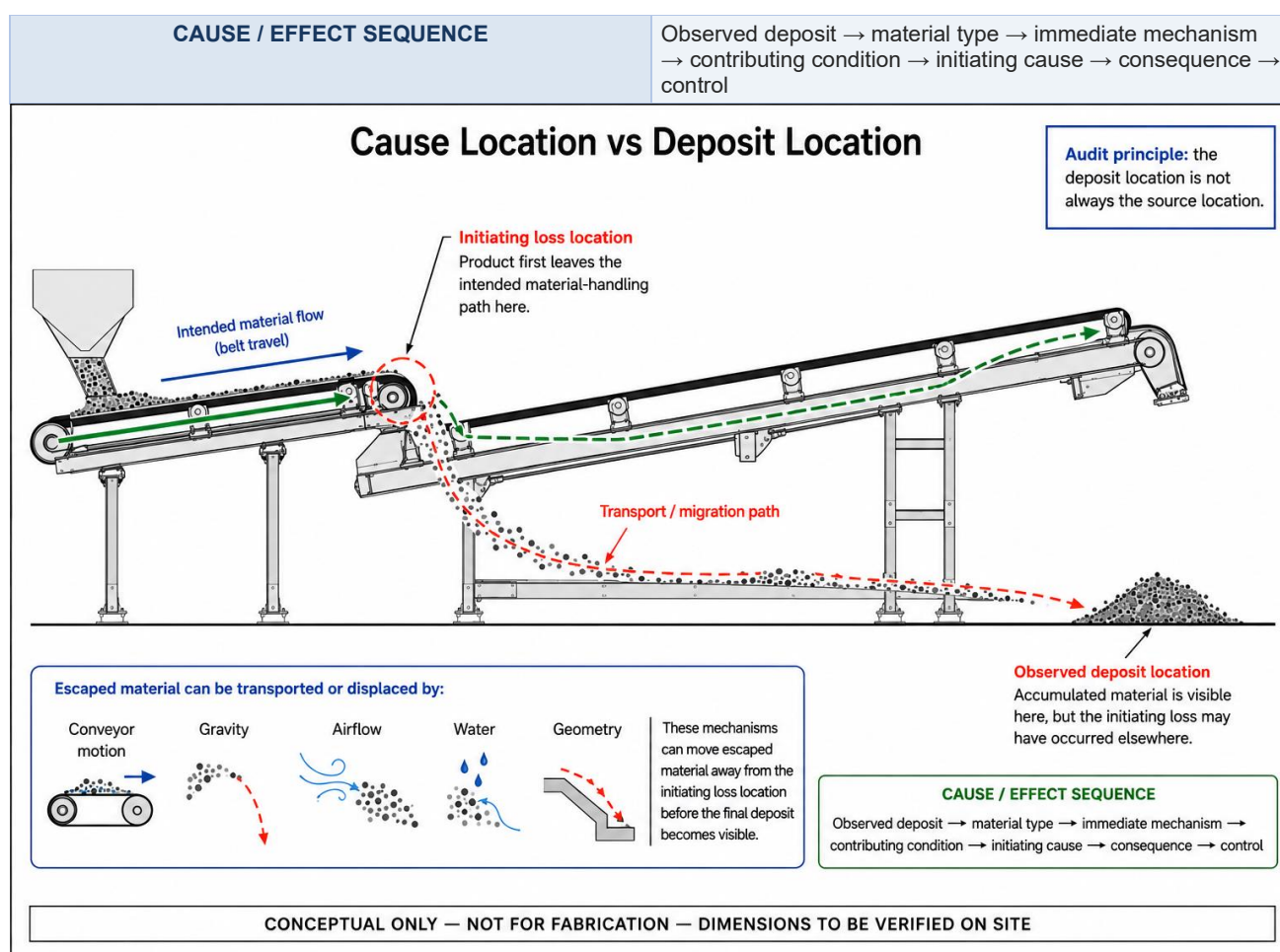


Figure 1 — Cause Location vs Deposit Location (CTD-004). Conceptual explanatory figure; not inspection evidence.

PART III — WHAT THE AUDIT FOUND

7 The Overall Spillage Picture

The evidence resolves into three recurring material-loss families and one event-driven operating mechanism: fines/carryback; coarse or mixed-product loss during transit; water-related escape, wet fines and slurry; and material removed during trip/restart or other operating interventions. These mechanisms can overlap at individual assets, but they do not all require the same physical control.

Table 7 — Material-loss mechanism families and control directions

Mechanism family	Typical evidence / significance	Control direction
Fines / carryback	Recurring deposits along return-side and downstream conveyor areas; cleaner condition and containment are recurring themes.	Improve cleaning function and retain removed fines inside the process.
Coarse / mixed product transit loss	Product outside the intended carrying path at W10, Sample Conveyors and other local positions.	Feed/trajectory, tracking and location-specific containment.
Water / wet fines / slurry	Wet deposits and confirmed active screen water escape.	Retain process water; identify local escape mechanism; manage secondary structural consequences.
Trip / restart unloading	Source-recorded manual clearing after trips/blockages.	Confirm controlled temporary handling, approved recovery/reintroduction and restart governance; distinguish event-driven deposits from chronic source-generated spillage.



W10-REF-002 — coarse/mixed product occupying restricted tunnel access.



CV01-REF-004 — wider view of wet-fines/slurry accumulation and standing water around the conveyor structure.



CV03-REF-002 — clean-condition comparator demonstrating that not every inspected conveyor was a significant spillage location.



UG5B-REF-006 — wider comparator supporting a localized rather than uniformly widespread U5B condition.

PART IV — FINES, CARRYBACK AND CLEANER CONTAINMENT

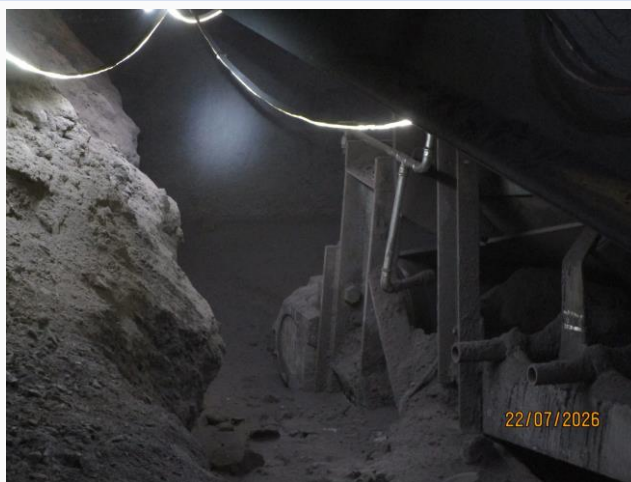
8 Fines Are Being Lost Progressively Through the Conveyor Ecosystem

Fines spillage was observed throughout the inspected conveyor ecosystem, with the extent varying considerably between locations. No inspected conveyor section was completely free of fines accumulation. The presence of fines was therefore system-wide, although the severity, local condition and immediate mechanism were not necessarily the same at every location. The repeated relationship between many deposits and return-belt/support-roller areas is consistent with carryback being the strongest recurring mechanism, particularly downstream of discharge locations. The photographs demonstrate representative accumulated consequences, but they do not establish that every individual deposit has the same sole origin.

The cross-system recurrence is important because the visible deposit may occur some distance downstream of the cleaner or discharge point where residual material should originally have been controlled. Material remaining on the return belt can detach progressively as the belt flexes and passes over rollers; return rollers/idlers can dislodge and redistribute additional adhered fines. Wet or sticky fines can be more persistent, although the cleaner-control philosophy applies to both wet and dry residual fines and the report does not assume one common water source or one identical immediate mechanism for all locations. The system-level objective is therefore to prevent residual fines from progressively leaving the process as the material moves from one conveyor and transfer stage to the next.



UGUC05_04_03_02_01-REF-010 — wider longitudinal accumulation supporting recurrence along the underground route.



UG5A-REF-006 — widespread return-side accumulation at U5A.



CV15-REF-004 — large fines deposits on grating/cross-bracing around the CV15 discharge region.



LoadoutStationSample1-REF-002 — substantial fines accumulation around the Sample Conveyor 1 tail/access area.

SYSTEM PATTERN

Carryback is a recurring system mechanism, not an isolated conveyor event. The report therefore focuses on cleaner function and containment at relevant discharges while retaining the evidence qualification applicable to each individual deposit.

9 Why Belt Cleaning Alone Is Not the Full Control Function

A belt cleaner needs to do more than simply remove material from the belt surface. For the cleaning function to be effective in practice, residual fines should be removed before they travel onto the return run, and the material removed by the cleaner should remain within the intended discharge or process route. Across the inspected system, several cleaner-related conditions were recorded that can reduce this effectiveness, including missing or worn blades, incomplete belt contact or coverage, reliance on manual adjustment, damaged or displaced support components, and cleaners positioned where removed material is not fully contained. The inspected conventional cleaner arrangements were also not understood to incorporate the proposed self-tensioning or wear-compensation principle. As the cleaner element wears, reduced compensation can allow cleaner-to-belt contact to deteriorate and more carryback to remain on the return belt. These specific cleaner deficiencies were not identified at every conveyor and should therefore be considered on a location-specific basis, even though fines accumulation itself was observed throughout the inspected conveyor route.

The maintenance context captured for the audit described a four-day operating / four-day maintenance cycle, with section artisans and supervisors understood to have maintenance responsibility. The audit did not review formal cleaner inspection standards, blade-replacement criteria or maintenance records, and it did not establish that one common cleaner design is used across the mine. The more practical approach is therefore to define and verify the required cleaning and containment function, while allowing the final installed arrangement to suit the local conveyor geometry, operating condition and maintenance requirements.

9.1 CV06 — Mechanical / Maintenance Failure Example

At CV06, the installed cleaner arrangement was not in a condition to provide the intended belt-cleaning function. The cleaner support structure remains in place, but the scraper blade elements are absent across the working belt width and a support bush/insert is visibly displaced. Poor housing design was recorded as a possible contributor to the bush displacement, with maintenance and replacement condition also relevant; however, the available photographs do not establish the root cause of the displacement or explain when and why the scraper blades were lost.

The immediate requirement is therefore to restore a complete and functional cleaner arrangement and verify the condition of its supporting components. The audit also puts forward improvements to the cleaner support, adjustment and wear-compensation arrangement so that the restored system is more maintainable and better able to retain its cleaning function through service; these proposed upgrades are developed further in the recommendations section.



CV06-REF-004 — displaced cleaner support bush/insert.



CV06-REF-005 — cleaner support structure present with scraper blades absent from the working belt width.

OBSERVATION	Physical cleaner defects are directly visible at CV06.
ENGINEERING ASSESSMENT	The missing scraper blades and displaced support component mean that the cleaner cannot provide the intended cleaning function in its inspected condition. Residual fines can therefore remain on the belt and continue onto the return run, contributing to carryback and subsequent fines spillage into the surrounding conveyor environment. Poor housing design was recorded as a possible contributor to the displaced bush/insert, but the available evidence does not establish the root cause or the history of the blade loss.
RECOMMENDED NEXT ACTION	Restore a complete functional cleaner arrangement; inspect and address the housing/support condition and retention, reinstate the required blade coverage, and verify cleaner contact, adjustment and maintainability. Incorporate the recommended support, adjustment and wear-compensation improvements where confirmed through the detailed maintenance/OEM review, and verify the completed arrangement under representative operation before close-out.

9.2 CV09 — Cleaner Removes Material, But Containment Still Fails

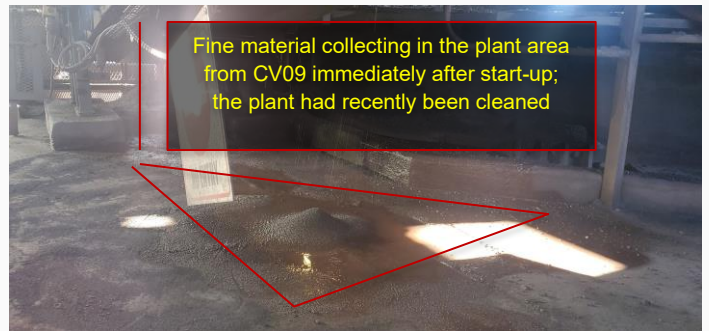
CV09 illustrates a different type of cleaner-related problem. The existing cleaner is positioned outside, or open to, the effective head-chute discharge containment arrangement. Although it removes some wet residual material from the belt, material successfully removed by the cleaner can fall directly into the surrounding plant area rather than remaining within the intended discharge path. At the same time, residual fines that remain on the belt can continue onto the return run and be progressively dislodged further downstream. This can contribute to local accumulation around CV09 and provides a credible pathway for wider downstream distribution of residual fines along the conveyor route, including into the lower Screens area where the conveyor route interfaces with the screening plant. The available evidence does not attribute every downstream fines deposit specifically to CV09.

Active wet-material fallout was recorded during operation, while the surrounding deposits provide visible evidence of the consequence of material not being fully retained within the process. CV09 is therefore a particularly useful location for demonstrating the difference between simply cleaning the belt and controlling what happens to the material after it has been removed. Because both the cleaner arrangement and the resulting fines-loss condition are clearly represented at this location, CV09 also provides a particularly suitable pilot location for proving an improved cleaner-and-containment approach before considering wider application elsewhere in the system.

The proposed functional objective is to remove residual fines before the belt enters the return run, with the cleaner acting while the belt is positively supported by the head pulley and with the cleaning stages located inside the existing head/discharge chute. In this arrangement, material removed by the cleaner can fall naturally into the existing combined-product discharge flow instead of escaping into the surrounding plant or requiring a separate fines-recovery chute. The audit defines this required function, while the final cleaner type, mounting arrangement, chute interfaces, access provisions and other detailed geometry remain subject to verified site dimensions and OEM/responsible-design development.



CV09-REF-003 — principal still showing the cleaner/open local arrangement.



CV09-REF-002 — wet-fines/slurry consequence below the discharge/cleaner region.



CV09-REF-005 — wider fines/material accumulation around the structure, supporting the broader consequence of material escaping and being redistributed outside the intended process path.



CV09-REF-008 — representative frame from inspection video showing contaminated cleaner/chute geometry relevant to the containment-boundary issue and pilot concept

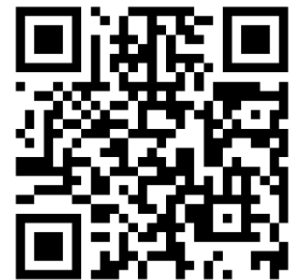


[View Video - CV09-REF-008](#)

Electronic copy: click the View Video link.
Printed copy: scan the QR code to view on mobile.



CV09-REF-006 — representative frame from inspection video showing active wet-material fallout during operation.



[View Video - CV09-REF-006](#)

Electronic copy: click the View Video link.
Printed copy: scan the QR code to view on mobile.

OBSERVATION	Wet fines/slurry and surrounding fines accumulation are present at CV09, and active wet-material fallout was recorded during operation. The evidence shows both the local cleaner/discharge arrangement and the consequence of material being lost outside the intended process path.
ENGINEERING ASSESSMENT	The existing cleaner is capable of removing some material from the belt, but its open relationship to the effective head-chute containment means that successfully removed wet fines can become plant-area spillage rather than remaining within the discharge process. Residual material can also remain on the belt and continue onto the return run, where it may be dislodged further downstream and contribute to fines accumulation in other areas of the plant. CV09 therefore demonstrates that effective belt cleaning and containment of the removed material need to be treated as one connected control function. Final cleaner geometry, component selection and detailed installation requirements remain subject to site verification and OEM/responsible-design development.
CONTROL PHILOSOPHY	Do not merely clean the belt. Keep the fines inside the process.
RECOMMENDED NEXT ACTION	Develop and trial the CV09 cleaner-and-containment concept from verified site geometry. The preferred arrangement is pulley-supported cleaning located inside the existing head/discharge chute, with two cleaning stages where practicable, effective wear compensation/self-tensioning, positive travel limitation and practical maintenance access. The completed pilot should be assessed under representative operating conditions to confirm that downstream carryback is materially reduced, cleaner-removed fines remain within the existing discharge flow and the arrangement remains maintainable without introducing unacceptable belt, cleaner, access or guarding problems.

9.3 What CV09 Means Beyond CV09

Taken together, CV06 and CV09 illustrate why the wider improvement programme should focus on what the cleaner system needs to achieve, rather than assuming that one identical scraper arrangement will suit every conveyor. At CV06, the immediate problem is the loss of basic cleaner function through missing blade elements and a displaced support component. At CV09, some material is removed from the belt, but the local arrangement does not adequately retain that material within the intended discharge process. These are different local conditions, but both ultimately affect the same objective: removing residual fines effectively and keeping those fines inside the process.

A wider cleaner-control approach should therefore be based on a common functional standard — effective belt cleaning, reliable contact as the cleaner wears, adequate mechanical support and travel control, practical maintenance access, retention of removed material within the intended process route, and verification that downstream carryback has been reduced under representative operation. Once the CV09 pilot has demonstrated that function, the relevant head/discharge locations across the inspected conveyor ecosystem should be systematically reviewed against the same standard. Where the local geometry, operating duty and maintenance requirements confirm applicability, the cleaner-and-containment function should then be progressively implemented or adapted. The intention is to standardize the required function rather than duplicate one identical cleaner arrangement at every conveyor.

U4 remains a location-specific exception because its cleaner and discharge geometry differ materially from the more conventional arrangements considered for wider application. Any departure from the common functional standard should therefore be based on the verified local geometry and operating requirement rather than assumed in advance.

10 Cleaner / Fines-Control Philosophy and CV09 Pilot

The findings at CV09 provide a practical opportunity to test an improved cleaner-and-containment approach because the existing cleaner arrangement, the active wet-material fallout, the surrounding fines accumulation and the intended destination of cleaner-removed material can all be related at the same location. CV09 also sits within the process route that carries fines separated during washing to the downstream fines collection/handling point. Rather than treating CV09 only as an isolated maintenance problem, it can therefore be used as a pilot location to demonstrate the wider end-to-end control objective: remove residual fines before the return run, keep cleaner-removed fines within the process, and carry the fines stream forward to the point at which it is intentionally collected and handled.

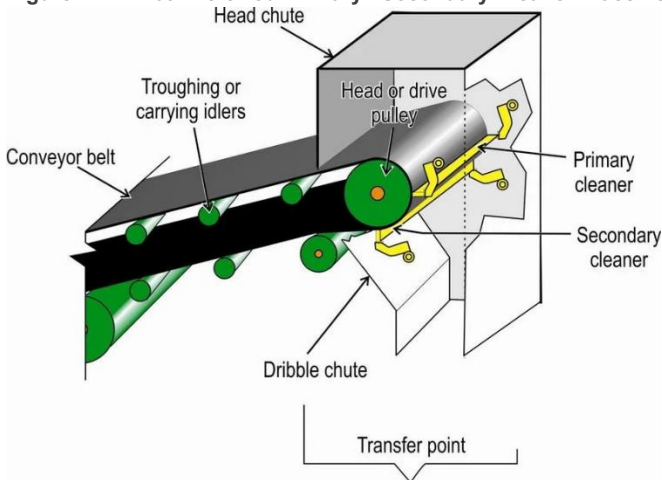
The proposed approach is based on a simple functional objective: remove residual fines before they enter the return run and keep the material removed by the cleaner within the existing discharge process. To achieve this, the cleaner should act while the belt is positively supported by the head pulley, with the cleaning stages positioned within the existing head/discharge-chute arrangement so that removed fines can fall naturally back into the combined-product discharge flow. The proposed cleaner arrangement should also incorporate a spring-loaded/self-tensioning mechanism that provides controlled wear compensation as the cleaner element wears. This helps maintain positive, controlled cleaner-to-belt contact through service and reduces reliance on repeated manual readjustment to compensate for normal wear. A two-stage cleaner arrangement is preferred where the verified site geometry allows it. Where practical constraints prevent two effective stages from being installed, one effective pulley-supported stage may be considered as the constrained minimum, subject to the final OEM/responsible-design assessment. If the pilot demonstrates the required performance, the function should become the reference standard against which the remaining relevant conveyor head/discharge locations are reviewed.

Indicative supplier reference: [Tru-Trac Rollers \(Pty\) Ltd](#) is identified as one suitable supplier for consideration during the detailed design and procurement of the proposed belt-cleaner arrangement. Tru-Trac manufactures primary and secondary belt-cleaner/scraper systems, including tensioned arrangements intended to maintain effective cleaner contact through blade wear. This supplier reference is indicative only and does not prescribe a specific Tru-Trac product, cleaner model or sole-source procurement route. South32 may select Tru-Trac or any other preferred supplier/OEM capable of meeting the functional, site, safety, maintainability and detailed-design requirements defined in this report. Final cleaner selection, stage arrangement, blade material, tensioning characteristics, contact forces, supports and installation geometry remain subject to verified site conditions and the responsible detailed-design process.

Conceptual arrangement

The proposed CV09 philosophy is illustrated by two complementary drawings:

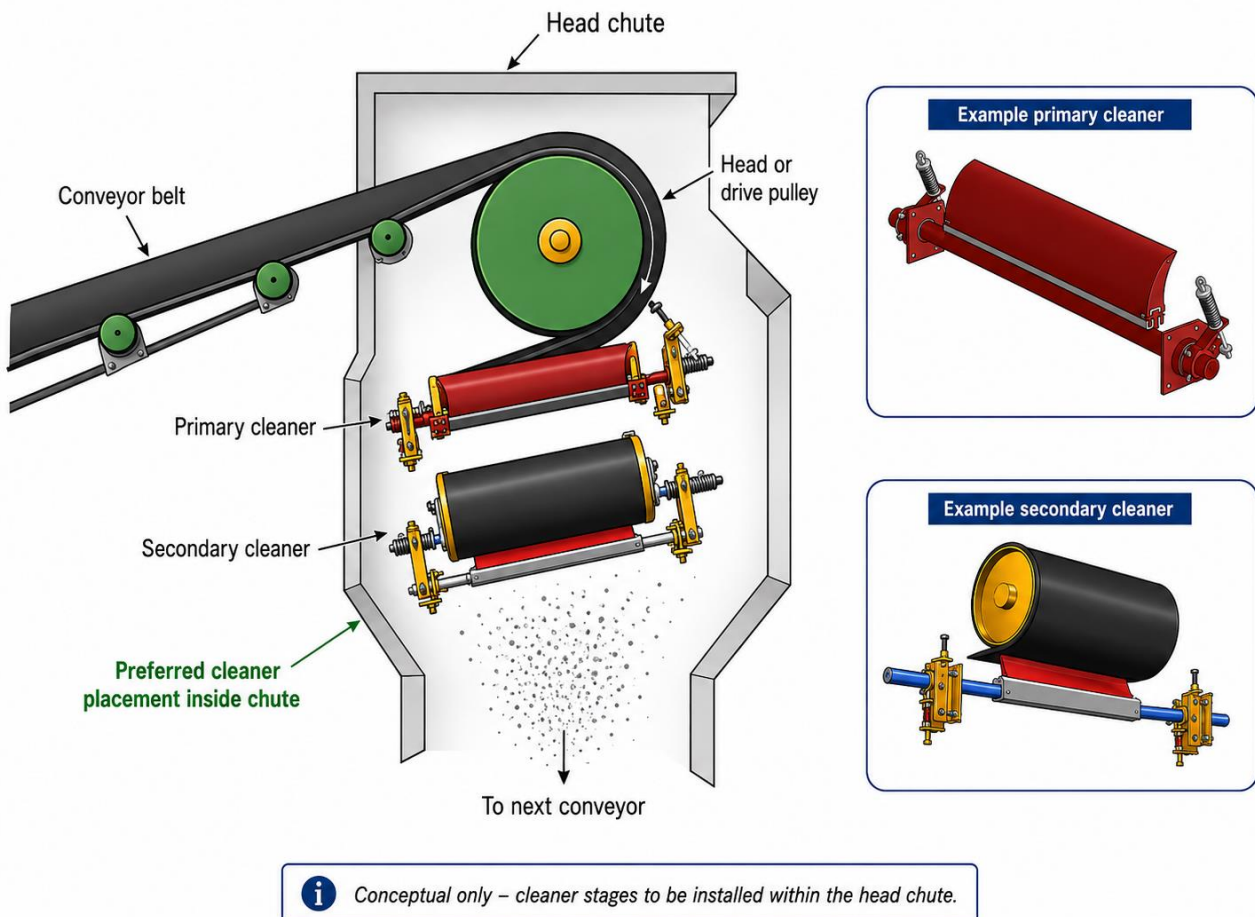
Figure 2 — CV09 Preferred Primary / Secondary Cleaner Placement and Orientation (CTD-005 / DRAW-005)



This figure illustrates the typical relationship and orientation of primary and secondary cleaner stages relative to the head pulley. Its purpose is to help explain **where the cleaning action should take place** while the belt remains positively supported. It is a placement/orientation reference and does not prescribe the final cleaner make, model or CV09 installation geometry.

Dust Control Handbook for Industrial Minerals Mining and Processing, RI 9689 (2012)

Figure 3 — CV09 Cleaners Inside Head-Chute Containment Principle (CTD-006 / DRAW-006)



This figure illustrates the complementary containment principle: the cleaner stages are positioned **inside the head/discharge chute** so that fines removed from the belt remain within the existing product flow rather than being discharged into the surrounding plant area.

The two drawings are intended to be read together. DRAW-005 is an external/reference placement illustration showing where and how the cleaner stages typically act relative to the head pulley. DRAW-006 is the project conceptual containment figure showing that the cleaner stages should sit within the head/discharge chute so that removed material remains inside the process. Neither drawing defines the final CV09 hardware or fabrication geometry; final dimensions, mounting arrangements, cleaner technology and site interfaces remain subject to verification and detailed design.

The drawings principally illustrate cleaner placement and containment. The spring-loaded/self-tensioning wear-compensation arrangement is an additional functional requirement of the proposed cleaner philosophy. Its purpose is to help maintain positive, controlled cleaner-to-belt contact as the cleaner element wears, while reducing reliance on repeated manual readjustment. The final spring arrangement, contact force, adjustment range and associated mechanical details remain subject to OEM/responsible detailed design.

Functional requirements for the pilot arrangement

The detailed CV09 design should be developed around the following functional requirements:

- Locate the cleaner contact where the belt is supported by the head pulley so that the belt remains mechanically stable through the cleaning zone.
- Provide effective cleaning across the required belt width while maintaining suitable cleaner-to-belt contact as the wear elements reduce through service.
- Provide a spring-loaded/self-tensioning wear-compensation mechanism that maintains positive, controlled cleaner-to-belt contact as the cleaner element wears through service. The arrangement should reduce the need for repeated manual adjustment while still allowing appropriate inspection and setting, and should include a positive mechanical travel/wear limit to prevent uncontrolled over-travel toward the belt.
- Keep the cleaner stages within the head/discharge-chute containment so that material removed from the belt falls naturally into the existing combined-product discharge flow rather than creating a separate fines-spillage point.
- Provide practical access for inspection, adjustment, cleaner-element replacement, guarding and clean-out.
- Confirm the final cleaner type, materials, contact arrangement, forces, supports, movement, structural attachments, dimensions and tolerances through the OEM/responsible detailed-design process.

STANDARDISE THE FUNCTION

The intended long-term position is not that every conveyor receives an identical cleaner installation. Rather, each relevant head/discharge location should provide the same required function: effective removal of residual fines before the return run, reliable cleaner contact through service, and retention of the removed fines within the intended downstream process route. The CV09 pilot is intended to demonstrate that functional standard. Once demonstrated, the remaining relevant conveyor locations should be systematically reviewed and progressively adapted against the same standard, with the physical arrangement developed to suit local belt condition, chute geometry, operating duty, maintenance access and available installation space. U4 remains a location-specific exception because its cleaner and discharge arrangement differs materially from the more conventional conveyor geometry considered for wider application.

Pilot verification

The value of the CV09 pilot should be judged by its performance over representative operation rather than simply by how clean the area appears immediately after installation. The as-found condition should first provide a practical baseline against which the modified arrangement can be compared.

Successful performance should demonstrate that downstream carryback is materially reduced relative to that baseline, cleaner-removed fines remain within the existing discharge process, and the spring-loaded/self-tensioning arrangement maintains effective cleaner-to-belt contact as the cleaner element wears without requiring excessive routine manual readjustment. The installation should also remain practical to inspect and maintain without introducing unacceptable belt wear, cleaner wear, material trapping, guarding conflicts or maintenance-access difficulties.

The resulting monitoring record should provide enough evidence to determine whether the same functional cleaner-and-containment philosophy is suitable for wider application at other conveyors, while still allowing each location to be designed around its own geometry and operating requirements.

PART V — COARSE / COMBINED-PRODUCT LOSS DURING TRANSIT

11 The Containment Problem Is Not Limited to Fines

The inspection also identified locations where coarse/lump or mixed product had moved outside the intended carrying area of the belt during transit. These conditions are different from return-side fines and carryback, and the appropriate response is therefore to focus on keeping the transported product on the belt until it reaches its intended transfer or discharge point. The conditions that influence this can vary from one location to another and may include feed quantity and variation, product-size distribution, loading position or trajectory, available belt carrying width, belt tracking, transfer geometry and the effectiveness of local side containment.

A practical approach is to first understand how and where the product is leaving the belt, then verify the relevant feed, loading, tracking and conveyor geometry and address any confirmed primary contributors where practicable. Once those conditions have been considered, any remaining need for additional lateral containment can be established more confidently. Where product still has a tendency to move toward or beyond the carrying edge, belt-compatible side skirting or spill sealing can then be used to help retain it on the belt. The aim is to control the material before it leaves the carrying path, rather than relying primarily on a barrier positioned outside the conveyor to intercept product after it has already escaped.

12 W10 — Coarse Product Loss in a Confined Tunnel

W10 is significant because substantial coarse and mixed-product accumulation is present outside the intended carrying path of the conveyor within a restricted tunnel environment. The photographs clearly show the resulting accumulation and the limited access available around the conveyor, although the present evidence does not establish one single initiating cause for every deposit. The response should therefore begin by understanding the conditions that are allowing product to move toward or beyond the belt edge, rather than assuming that the surrounding accumulation is caused by one isolated defect.

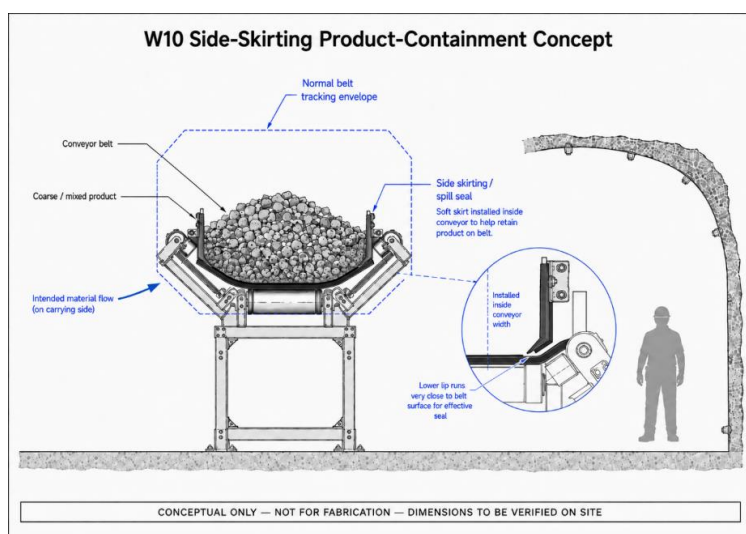
Relevant contributors such as loading position and distribution, feed condition, belt tracking and the local belt, trough and support geometry should be checked and corrected where a contributing condition is confirmed. In parallel with this verification, belt-compatible side skirting or spill sealing is the preferred engineered containment addition for W10 because substantial lateral coarse/mixed-product loss has already been demonstrated within the restricted tunnel environment. The verification work should therefore establish the corrected operating condition and final containment duty around which the skirting arrangement is designed, rather than determine whether the skirting concept is considered at all.

The proposed skirting is intended to act directly at the conveyor carrying edge, along the conveyor sides and inside the carrying width, so that laterally moving product is retained on the belt before it can escape into the surrounding tunnel area. The skirt or seal should form a near-vertical containment surface along the troughed belt edge, with the lower sealing relationship selected so that normal belt movement and tracking can be accommodated without damaging rubbing, rigid belt guidance, excessive product trapping or inaccessible build-up.

Conceptual skirting arrangement

The proposed W10 containment philosophy is illustrated by two complementary drawings.

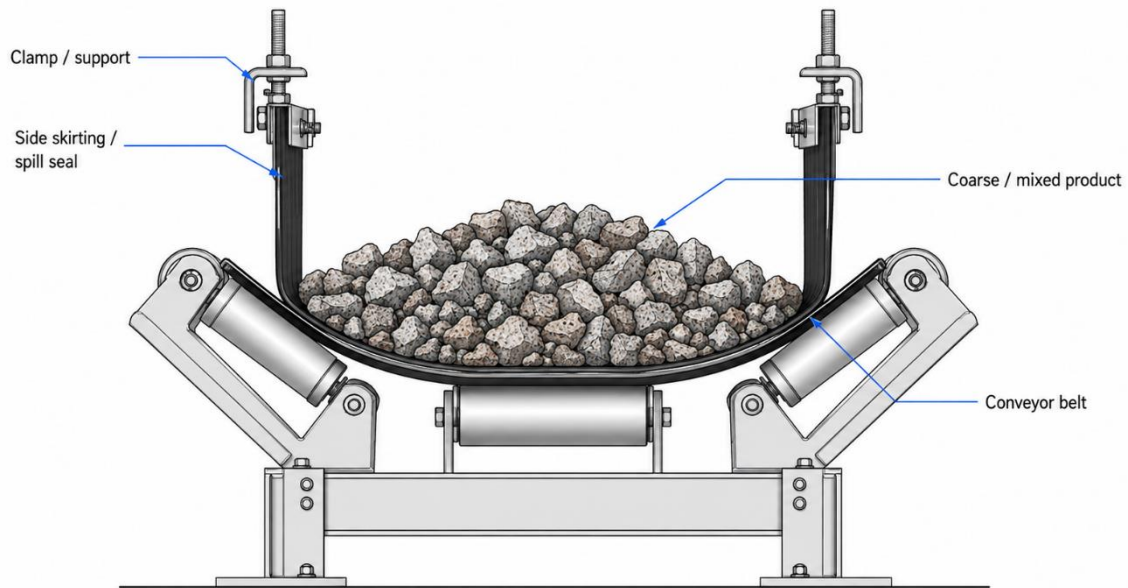
Figure 4 — W10 Side-Skirting / Product-Containment Concept (CTD-010 / DRAW-0010)



This drawing places the proposed skirting into the W10 tunnel context. Its purpose is to show where the skirting is intended to act relative to the conveyor, carried product and restricted surrounding access. The important point is that the containment element sits at the conveyor carrying edge and acts before product has escaped into the surrounding tunnel.

Figure 5 — Conveyor Side Skirting / Spill Seal Concept — Cross-Section (CTD-011 / DRAW-0011)

Conveyor Side Skirting / Spill Seal Concept (Cross-Section)



This drawing explains how the skirting principle works in cross-section. A flexible skirt or spill-seal element is supported along the conveyor side and extends downward on the inside of the carrying width, close to or in controlled belt-compatible contact with the troughed belt surface as determined by the selected system. In this way, the skirt provides a longitudinal containment boundary that helps retain coarse and mixed product on the belt.

The two drawings should therefore be read together: DRAW-0010 shows where the skirting is used in the W10 conveyor/tunnel arrangement, while DRAW-0011 explains the physical side-skirt/spill-seal mechanism. Neither drawing defines the final OEM product, skirt material, dimensions, contact pressure, clamp arrangement or installation geometry. Those details remain subject to verified site geometry and OEM/responsible detailed design. This is also the controlling interpretation in the current conceptual-figure register.

Wider applicability of the principle

W10 provides a preferred location for demonstrating the side-skirt/spill-seal containment principle because substantial lateral product loss has already been recorded and the restricted tunnel environment makes escaped product particularly difficult to access, recover and manage. The completed W10 arrangement should therefore be verified under representative operation as the principal demonstration of the proposed skirting concept.

Once satisfactory performance has been demonstrated, the same functional principle should be considered and adapted at other underground or surface conveyor locations where inspection or operating experience identifies a comparable lateral-product-loss or belt-edge containment requirement and the local mechanism, belt condition and conveyor geometry confirm that skirting is suitable. This wider applicability is not limited to confined areas, although restricted-access or spatially constrained underground conveyor sections are particularly relevant potential applications because preventing product escape can materially reduce difficult housekeeping and access consequences. The intention is to apply the proven containment principle where it is technically appropriate, not to reproduce the W10 hardware or geometry unchanged at every conveyor.

The same side-skirt/spill-seal principle is also proposed for Sample Conveyors 2 and 3. At those conveyors, however, implementation should follow quantified assessment and correction or adjustment of the feed quantity/variation, product-size distribution, loading position, usable carrying geometry and belt-position/tracking or alignment condition. The corrected operating condition should then be used to define the final skirting extent, geometry and containment duty.



W10-REF-010 — long tunnel view establishing the confined topology and restricted access.



W10-REF-002 — principal coarse/mixed-product consequence in the access route.



W10-REF-006 — product distributed below the conveyor around components, explaining local geometry.



W10-REF-009 — product immediately beside support steel; useful context for local side-containment feasibility.

OBSERVATION	Coarse and mixed product has moved outside the intended carrying path and accumulated within the restricted W10 tunnel area. The photographs demonstrate both the scale of the accumulated consequence and the limited access available around the conveyor.
ENGINEERING ASSESSMENT	The demonstrated lateral coarse/mixed-product loss and restricted W10 tunnel environment support belt-compatible side skirting or spill sealing positioned within the carrying width as the preferred engineered containment addition. The arrangement should act directly at the carrying edge to retain laterally moving product on the belt, while verification and correction of loading, tracking and local geometry contributors establish the operating condition around which the final skirt/seal arrangement is designed.
RECOMMENDED NEXT ACTION	Verify representative loading and belt tracking together with the local belt, trough and support geometry, and correct any confirmed primary contributors. Using the verified and corrected operating condition, develop and install the preferred belt-compatible side-skirt/spill-seal arrangement suited to the W10 geometry, with the final extent, sealing relationship, support and wear-adjustment requirements established through the selected OEM/responsible design. The completed arrangement should then be assessed under representative operation to confirm that coarse/mixed-product escape is materially reduced, normal belt movement and tracking are accommodated, the skirt remains belt-friendly and maintainable, and clean-out and access are not adversely affected. If the principle performs successfully, the same functional containment approach should be considered and adapted at other appropriate underground or surface conveyor locations where a comparable lateral-containment requirement is identified and local conditions confirm suitability, including but not limited to restricted-access sections and the Sample Conveyors following completion of their prerequisite feed/product/loading/tracking corrections.

13 Sample Conveyors — Product Size, Feed and Limited Carrying Width

Sample Conveyors 2 and 3 illustrate how several operating and geometric conditions can interact to reduce the carrying margin available for coarse and mixed product. Feed quantity and variation, product-size distribution, loading position, usable belt width and the lateral position of the belt can all influence how effectively the product remains within the intended carrying area.

The available photographs do not establish that either conveyor was operating above its rated or design throughput capacity. Product can nevertheless be lost below rated throughput where relatively large pieces, uneven or off-centre loading, limited carrying width or lateral belt displacement reduce the space available for the material to remain securely contained on the belt.

A practical approach is therefore to establish the intended operating basis and quantify the actual feed and product condition, then assess how the material is being loaded and how the belt is tracking under representative operation. Where feed, product sizing, loading, tracking, alignment or local interference is confirmed as a contributing condition, it should be corrected or adjusted before the proposed side-skirt/spill-seal arrangement is finalized. This allows the same skirting principle demonstrated at W10 to be developed around the corrected Sample Conveyor operating condition rather than being used to mask or compensate for a primary contributor that has not yet been understood or rectified.

PRACTICAL ASSESSMENT AND CONTROL SEQUENCE	1. Establish the intended/design basis and representative feed quantity/variation. 2. Verify representative product-size distribution and loading position/centering. 3. Quantify the belt-position/tracking condition and investigate alignment or local interference where indicated. 4. Correct confirmed contributors and reassess the usable carrying geometry and remaining feed/product controls. 5. Define and install/adapt belt-compatible side skirting or spill sealing to suit the corrected operating condition and verified lateral-containment duty. 6. Verify the combined feed → product size → loading → tracking → containment performance under representative operation.
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13.1 Sample Conveyor 2

At Sample Conveyor 2, the photographs show coarse/lump product occupying a substantial proportion of the available belt carrying width, together with material outside the intended carrying path. The REF-004 and REF-005 views also support a captured lateral belt-offset condition. The available evidence does not establish the magnitude, persistence or initiating cause of that offset, nor does it establish that the conveyor was operating above a quantitative design-capacity limit. Feed quantity and variation, product-size distribution, loading position, belt position and the congested entry/guide area should therefore be considered as interacting conditions requiring representative operating verification.



LoadoutStationSample2-REF-004 — Coarse/lump product occupying a substantial proportion of the available Sample Conveyor 2 carrying width. When read with REF-005, the belt/roller relationship also supports the captured lateral belt-position condition.



LoadoutStationSample2-REF-001 — Wider view of coarse/lump-product accumulation around Sample Conveyor 2, showing the surrounding consequence of local product loss.



LoadoutStationSample2-REF-005 — Complementary view of the Sample Conveyor 2 belt/roller relationship, with a substantial portion of the roller exposed. Read with REF-004, this supports a captured lateral belt offset; the magnitude remains unmeasured.



LoadoutStationSample2-REF-007 — Representative loading and product-size view showing coarse product relative to the limited Sample Conveyor 2 carrying width.



LoadoutStationSample2-REF-006 — Sample Conveyor 2 entry/guide area showing accumulated and displaced product around the local covers and outside the intended carrying path. The view provides context for the interacting feed, product-size, belt-position and containment conditions.

The recommended response is therefore to establish the intended Sample Conveyor duty and existing feed-control arrangement, quantify representative feed and product size, assess loading centering/distribution, quantify the captured belt-position condition and inspect the congested entry/guide area for contributing interference or accumulated product. Confirmed feed, product-size, loading, tracking/alignment or interference contributors should be corrected or adjusted as required and acceptable belt positioning demonstrated under representative operation. The proposed belt-compatible side-skirt/spill-seal arrangement should then be developed and installed around that corrected operating condition, with its final extent and containment duty determined from the verified usable carrying geometry.

13.2 Sample Conveyor 3

Sample Conveyor 3 shows a similar relationship between relatively large product and limited available carrying width, together with a visually substantial lateral belt offset in the REF-006 and REF-007 views. The magnitude and persistence of the offset have not been measured, and the available evidence does not establish the mechanical condition responsible for it.

The appropriate next step is therefore to assess representative feed, product size and loading position together with the belt-position/tracking condition. Quantitative tracking/alignment checks and inspection of relevant mechanical or interference contributors can then be used to identify any primary condition requiring correction before the proposed side-skirt/spill-seal arrangement is finalized around the corrected operating condition.



LoadoutStationSample3-REF-002 — Feed/loading view showing coarse product relative to the limited Sample Conveyor 3 carrying width, with material also visible outside the intended carrying path.



LoadoutStationSample3-REF-003 — Supporting view showing smaller fragments retained on the belt while larger pieces are visible outside the intended carrying path.



LoadoutStationSample3-REF-004 — Wider relationship view showing product size, belt position, available carrying margin and material outside the intended carrying path within the same local area.



LoadoutStationSample3-REF-005 — Additional view showing large coarse product occupying much of the available Sample Conveyor 3 carrying width.



LoadoutStationSample3-REF-006 — Sample Conveyor 3 belt/roller view showing a substantial portion of the roller exposed and supporting a visually substantial captured lateral belt offset.



LoadoutStationSample3-REF-007 — Complementary Sample Conveyor 3 view showing the captured lateral belt displacement. The magnitude of the offset remains unmeasured.



LoadoutStationSample3-REF-012 — Wider view of mixed-size product accumulation on the surrounding grating/access area adjacent to Sample Conveyor 3.

OBSERVATION	Coarse product occupies a significant proportion of the available carrying width and substantial surrounding product accumulation is visible. The REF-006/REF-007 pair also supports a visually substantial captured lateral belt offset, although its magnitude has not been measured.
ENGINEERING ASSESSMENT	The photographs support a lateral belt-position concern but do not establish its dimension, persistence or mechanical cause. Feed condition, product-size distribution, loading position and belt tracking may interact and should therefore be assessed together before one condition is assigned as the primary cause of the observed product loss.
RECOMMENDED NEXT ACTION	Establish the intended design/feed basis and quantify representative feed quantity/variation, product-size distribution and loading position. Quantify the belt-position/tracking condition and inspect the relevant alignment, tension, support and interference contributors. Correct or adjust confirmed contributors and verify acceptable centred operation under representative conditions. The proposed belt-compatible side-skirt/spill-seal arrangement should then be developed and installed around the corrected feed, loading and belt-position condition, with its final extent and containment duty determined from the verified carrying geometry. The completed combined arrangement should then be assessed under representative operating conditions.

Table 8 — Sample Conveyor 2 and Sample Conveyor 3 comparison

Question	Sample Conveyor 2	Sample Conveyor 3
Product-size / carrying-width concern	Supported	Supported
Captured lateral belt-offset condition	Supported by REF-004/REF-005; magnitude, persistence and cause remain unverified	Visually substantial offset supported by REF-006/REF-007; magnitude, persistence and cause remain unverified
Offset magnitude measured	No — measurement required	No — measurement required
Primary investigation focus	Design/feed/product/loading + quantify belt position/tracking + inspect entry/guide congestion	Feed/product/loading + quantify tracking/alignment and mechanical contributors
Secondary containment	Apply belt-compatible side skirting/spill sealing after confirmed feed/product/loading/tracking contributors have been quantified and corrected or adjusted; final extent and duty to suit the verified operating condition	Apply belt-compatible side skirting/spill sealing after confirmed feed/product/loading/tracking contributors have been quantified and corrected or adjusted; final extent and duty to suit the verified operating condition

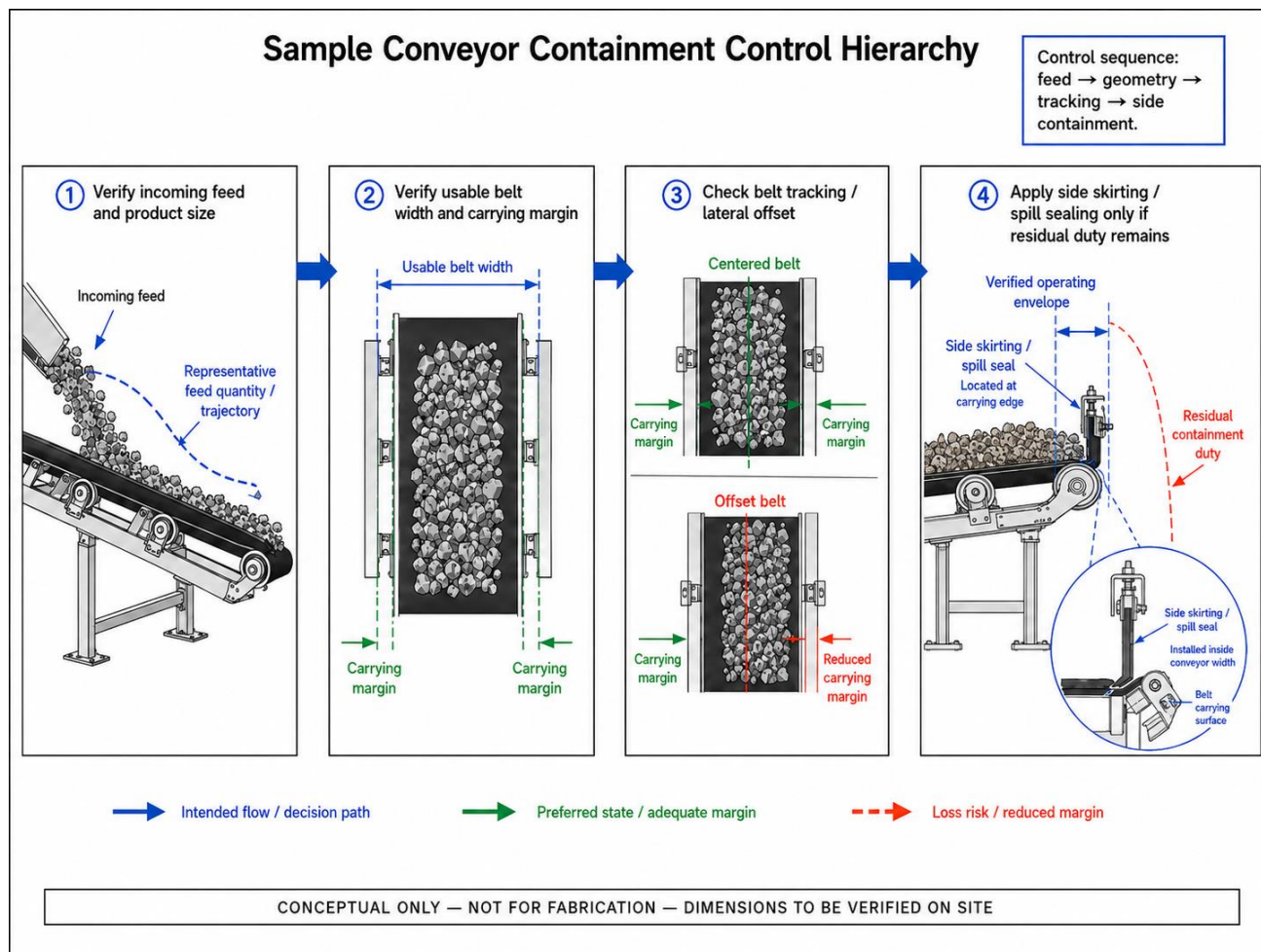


Figure 6 — Sample Conveyor Containment-Control Hierarchy (CTD-003). Conceptual only; not for fabrication; site geometry to be verified.

PART VI — ALIGNMENT AND LOCAL PRODUCT-LOSS MECHANISMS

14 Belt Tracking and Material Containment

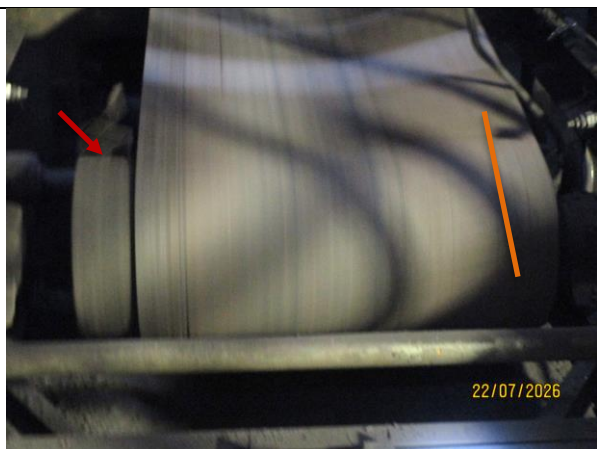
Belt tracking influences how much of the available carrying width can be used effectively to retain product on the belt. When the belt remains reasonably centred within the supporting structure, the available carrying margin is more evenly distributed to either side. Where the belt moves laterally, the carrying margin on one side is reduced, which can make it easier for coarse or mixed product to approach or move beyond the belt edge. Belt tracking should therefore be considered not only as a mechanical condition, but also as one of the factors that can influence material containment.

15 U4 Tripper Car — Belt Tracking and One-Sided Product Loss

U4 provides a useful example of how belt position can influence where coarse product is lost from a conveyor. The inspection recorded a concern with the tail-pulley/alignment relationship, while the photographs show the belt displaced laterally relative to the supporting geometry and larger/coarse product accumulated predominantly along one side of the tripper-car route. Taken together, these observations support a relationship between the tracking condition and the directional product-loss pattern.

The photographs do not provide a dimensional alignment measurement or establish the complete mechanical reason for the belt displacement. The tail-pulley relationship, idlers, tension/training arrangement, tripper geometry and any local interference should therefore be checked using fixed references and under representative operating conditions. A separate accumulation of fines is also present around the head-pulley/tripper discharge area and should be assessed independently from the coarse-product tracking condition.

During the inspection, product loss from one side of the belt was also reported when the tripper was positioned further along its travel, which makes full-travel operating verification particularly important.



UGU4-REF-008 — Tail-end belt/pulley view supporting the alignment concern recorded during the inspection. The relationship is visually apparent but was not dimensionally measured.



UGU4-REF-009 — Longitudinal view showing the belt displaced laterally relative to the carrying-idler/support geometry.



UGU4-REF-007 — Larger/coarse product accumulated predominantly along one side of the U4 Tripper Car conveyor

UGU4-REF-004 — Wider view showing the lateral belt-position condition extending toward the tripper-car region.

route. The photograph shows the accumulated consequence rather than the instant at which material left the belt.

The one-sided coarse-product accumulation should be distinguished from the substantial fines deposits also present around U4. REF-005 and REF-006 show this second condition. Although both forms of accumulation occur within the same tripper-car installation, they need not share the same immediate mechanism and should therefore be assessed separately.



UGU4-REF-005 — Substantial fines/product accumulation beside the U4 Tripper Car, providing context for the separate fines/carryback condition.



UGU4-REF-006 — Large fines accumulation in the head-pulley/tripper discharge area, showing the extent of the separate fines-related condition at U4.

OBSERVATION	A lateral belt-position/tracking condition is visible at U4, together with larger/coarse product accumulated predominantly along one side of the tripper-car route. A separate substantial fines-accumulation condition is also present around the head-pulley/tripper discharge area.
ENGINEERING ASSESSMENT	Taken together, the alignment context, visible belt displacement and directional coarse-product accumulation support a relationship between belt tracking and the observed one-sided product loss. The available evidence does not establish the dimensional alignment error or the complete mechanical cause of the tracking condition. The separate fines accumulation should be investigated on its own merits, including the local cleaner/discharge arrangement.
RECOMMENDED NEXT ACTION	Record the as-found belt and tail-pulley relationship using fixed, repeatable references and inspect the relevant pulley, idler, tension/training, tripper geometry and local-interference conditions. Where a contributing mechanical condition is confirmed, correct it and then observe the belt through the normal tripper travel under representative loading to confirm that an acceptable centred relationship is maintained. The one-sided coarse-product loss and the separate fines condition should then be reassessed independently to determine whether any further containment or cleaner-related action is required.

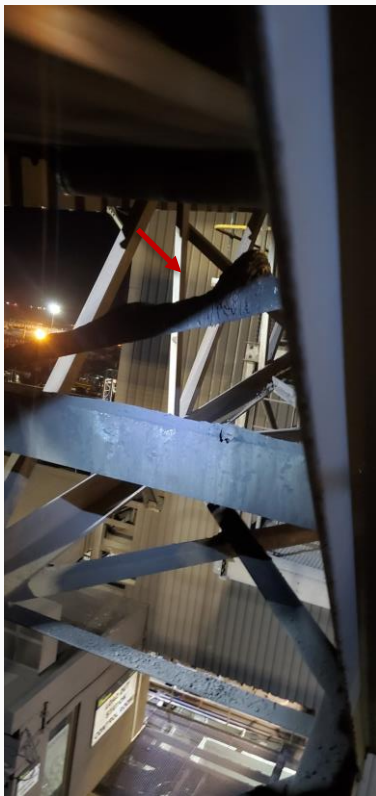
16 L3 — Alignment Concern and Separate Fines Accumulation

L3 presents a different evidence relationship from that observed at U4. At the tail end, the available photograph shows an asymmetric belt/pulley relationship that is consistent with an alignment or tracking concern. The magnitude of this condition has not been measured and should therefore be confirmed through dimensional checks and representative operating observation.

Fines and material accumulation are also visible at several locations around the L3 conveyor and supporting structure. These deposits establish a separate material-loss condition; however, the available evidence does not show that the tail-end alignment condition is responsible for the observed fines accumulation or for significant coarse-product loss. The two conditions should therefore be assessed separately, with the alignment/tracking relationship quantified first and the fines condition then reassessed in the context of the verified operating condition.



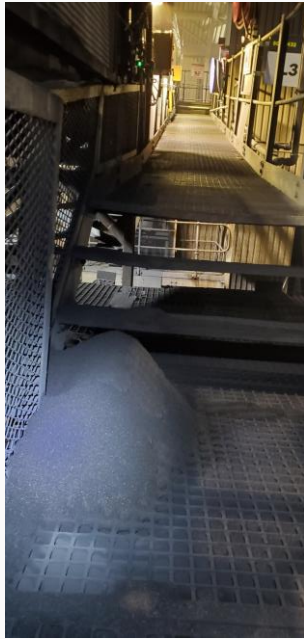
LoadoutStationL3-REF-001 — Tail-end view showing an asymmetric belt/pulley relationship consistent with an alignment or tracking concern. The magnitude of the condition remains unmeasured.



LoadoutStationL3-REF-004 — Additional view showing fines/material distributed around the local conveyor and support area. The photograph provides consequence context but does not establish the initiating source.



LoadoutStationL3-REF-007 — Close view showing substantial fine-material accumulation on the guarding and adjacent structure at L3.



LoadoutStationL3-REF-002 — Fines/material accumulation around the L3 conveyor and guarding area, illustrating the separate fines-spillage condition.



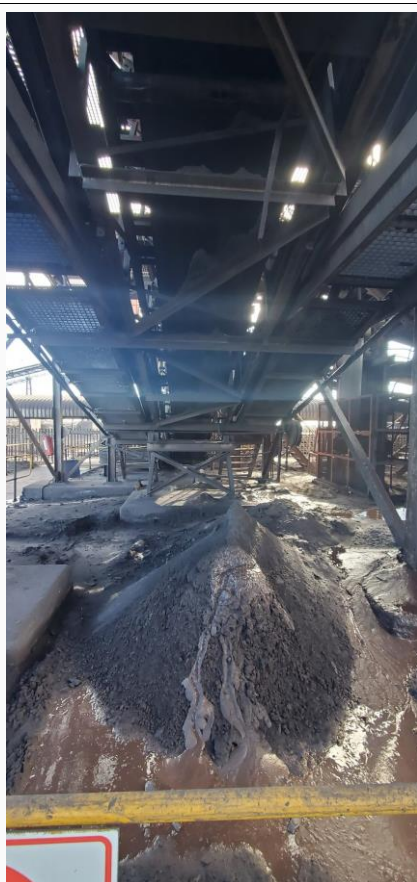
LoadoutStationL3-REF-008 — Supporting lower-area view showing fines and small-fragment accumulation beneath the conveyor.

PART VII — WATER, WET FINES AND STRUCTURAL CONSEQUENCES

17 Water, Wet Fines and Repeated Wetting Exposure

Water can change the way fines and deposited material behave once they leave the intended process route. Where process or cleaning water comes into contact with loose fines, the material can become wet or slurry-like and may be redistributed beyond the immediate area in which it was first released. This can increase housekeeping demand and expose nearby equipment and structures to repeated wetting.

The inspection record identifies several ways in which water may be introduced at different parts of the plant, including dust-suppression sprays, hose cleaning and high-pressure water used under some blockage or silo conditions. These are location-specific operating examples and should not be assumed to apply to every area where water or wet fines are visible. At CV01 and CV04, the photographs show wet-fines/slurry conditions, but the initiating water source has not been established from the available evidence. At the Screens, considered in the following section, operating video provides a stronger basis because active process-water escape was directly observed. Repeated wetting may be relevant to continued equipment and structural exposure, but it should not on its own be used to assign the cause of corrosion or deterioration.



CV01-REF-003 — Wet-fines/slurry accumulation with pooled water beneath the conveyor area. The photograph shows the wet condition, while the initiating water source remains unresolved.



CV04-REF-002 — Saturated wet-fines/slurry accumulation beneath CV04, with visible wet fallout/droplet evidence in the local area.



CV04-REF-005 — Wider view showing multiple fines accumulations beneath and around CV04, illustrating the broader distribution of material at this location.

18 Screens — Active Water Escape and Structural Condition

At the Screens, water forms part of the operating process, but the inspection also recorded water escaping beyond the intended screen/chute process route. The operating videos show active spray and runoff, with Screen-REF-017 providing useful visual localization of the external area from which the water appears to emerge. Because the equipment was operating during the inspection, the corresponding internal area could not be safely examined and the component or condition responsible for the escape remains unresolved. A planned shutdown inspection can therefore use the observed external location as a starting point to examine the corresponding internal area, identify the water-loss mechanism and determine the appropriate corrective action. Following any corrective work, the area should then be observed under representative operating conditions to confirm that the water remains within the intended process route.

The same area also shows substantial corrosion and deterioration, together with support members that contain visible cut-outs or notches. These conditions are important in their own right and should be considered separately from the water-loss investigation. The photographs establish the physical deterioration and altered-member geometry, but they do not establish why the openings were made, the remaining capacity of the affected members, or that the observed water escape is the sole cause of the corrosion. A competent structural assessment should therefore establish the role and load path of the affected members, record the relevant dimensions, section loss, cut/notch geometry and connection condition, and determine whether repair, strengthening, replacement or interim controls are required.

Corrosion / Deterioration



Screen-REF-008 — Wider deterioration view showing the broader extent of corrosion and section deterioration within the Screens structure.



Screen-REF-002 — Close view showing substantial local corrosion and deterioration within the Screens structure. The photograph establishes the visible condition but does not determine remaining structural capacity or its complete cause.



Screen-REF-003 — Wider view of the screen/chute structural area, providing context for the corrosion and deterioration shown more closely in REF-002.

Active Water Escape

	
Screen-REF-017 - Representative frame from the operating video showing the external area from which substantial water spray/runoff emerges. The view helps localize the escape point for follow-up shutdown inspection; the internal source remains unresolved. View Video - Screen-REF-017 <i>Electronic copy: click the View Video link. Printed copy: scan the QR code to view on mobile.</i> 	Screen-REF-016 — Representative frame from the operating video showing active process-water spray/escape from the screen/chute area. The video confirms active water loss but does not establish the internal defect or source mechanism. View Video - Screen-REF-016 <i>Electronic copy: click the View Video link. Printed copy: scan the QR code to view on mobile.</i> 

Support-Member Cut-Outs / Notches

 Screen-REF-009 — Wider view showing local cut-outs/notches in the affected support-member arrangement. The reason for the alterations and their effect on remaining structural capacity has not been established.	 Screen-REF-010 — Close view showing a substantial cut-out in an affected support member. The physical alteration is clearly visible; structural adequacy requires assessment by the responsible structural authority.
OBSERVATION	Active process-water escape is visible in the operating evidence. Substantial corrosion/deterioration and physical cut-outs/notches in support members are also visible within the Screens area. Although these conditions occur within the same operating environment, the available evidence does not establish that they arise from one common cause.
ENGINEERING ASSESSMENT	The operating evidence provides a useful external location from which the water-loss investigation can proceed, but the internal component or condition responsible for the escape remains to be established. Repeated wetting may increase the ongoing exposure of the surrounding structure, although the available evidence does not establish it as the sole cause of the observed corrosion. The support-member cut-outs/notches are physically evident, but their history, structural significance and remaining capacity require competent structural assessment.
RECOMMENDED NEXT ACTION	Use the externally identified water-escape location to guide a planned shutdown inspection of the corresponding internal screen/chute area. Identify the water-loss mechanism and correct the confirmed condition, then re-check the area under representative operating conditions. Separately, complete a competent structural assessment of the corroded and altered support members and implement the resulting repair, strengthening, replacement or interim-control requirements, followed by formal acceptance of the completed structural work.

19 L1 — Accumulation, Previous Belt Contact and Structural Deterioration

The L1 area shows several conditions that need to be considered together but not automatically attributed to one common cause. Material and fines have accumulated across the conveyor/access area and are also present on and around parts of the supporting structure. Local concrete deterioration includes material loss and exposed reinforcement, while separate steel support locations show substantial corrosion and section loss. Wear slots are also visible beside the belt path and are consistent with previous or intermittent belt-to-structure contact.

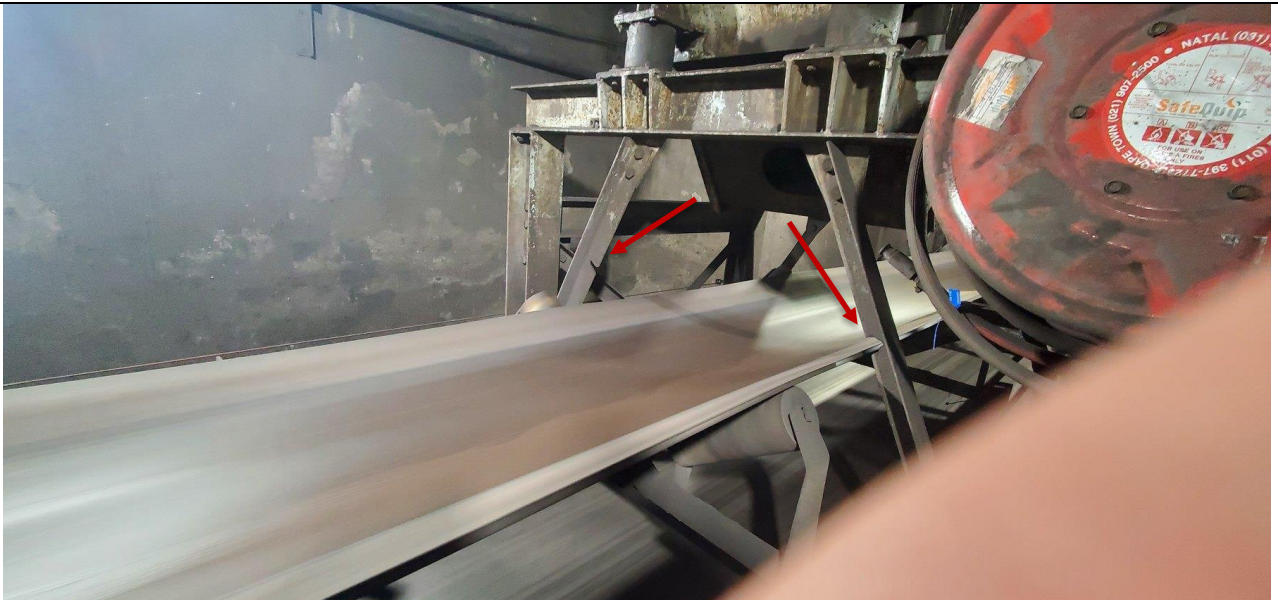
In the operating condition captured during the inspection, the belt was running clear of the affected supports and no obvious lateral offset, mistracking or contact was visible. The operating circumstances that produced the wear marks were not witnessed and therefore remain to be established. Follow-up should combine competent structural assessment of the deteriorated concrete and steel, measurement of the wear/contact geometry and representative tracking/clearance checks, together with measures to reduce recurring material, fines and moisture retention around any repaired structural interfaces. The photographs establish the visible condition but do not determine remaining structural capacity or prescribe the final repair arrangement.



LoadoutStationL1-REF-013 — Broad view showing extensive material and fines accumulation across the L1 conveyor/access area.



LoadoutStationL1-REF-017 — Elevated view showing material accumulation retained on and around the L1 structural members, complementing the broader floor/access-area accumulation shown in REF-013.



LoadoutStationL1-REF-007 — Wear slots adjacent to a belt that was running clear at the inspected condition. The wear is consistent with previous or intermittent belt-to-structure contact, while continuous present contact is not established.



LoadoutStationL1-REF-004 — Wider view showing the location and surrounding context of concrete deterioration at L1.



LoadoutStationL1-REF-005 — Close view showing substantial concrete material loss/spalling with exposed reinforcement. Remaining structural capacity cannot be determined from the photograph.



LoadoutStationL1-REF-008 — Local steel deterioration and section loss at one L1 conveyor support location.



LoadoutStationL1-REF-009 — Steel deterioration and section loss at a separate L1 support location, showing that the condition is not limited to the support illustrated in REF-008.

OBSERVATION	Extensive material/fines accumulation is present around L1, together with concrete deterioration and exposed reinforcement, local steel section loss at more than one support location, and wear consistent with previous/intermittent belt-to-structure contact. In the operating condition observed during the inspection, the belt was running clear of the affected supports and no obvious lateral offset, mistracking or contact was visible.
ENGINEERING ASSESSMENT	The wear slots are consistent with previous or intermittent belt contact, but the available evidence does not establish a continuous present rubbing condition or the mechanical circumstances that produced the wear. The visible concrete and steel deterioration requires structural assessment because remaining capacity cannot be determined photographically. Material and moisture accumulation around the structure may contribute to continued exposure and repeated accumulation around the affected interfaces, but the available evidence does not establish these conditions as the cause of every structural defect.
RECOMMENDED NEXT ACTION	Complete a competent structural assessment of the affected concrete and steel supports and implement the resulting approved repair, strengthening or replacement requirements. Record the wear/contact geometry and observe representative belt tracking and clearance to determine whether a current mechanical contributor remains present. Where a contributing condition is confirmed, correct it and verify the as-left operating clearance. Recurring material, fines and moisture retention around repaired structural interfaces should also be reduced and the repaired condition monitored.

PART VIII — EVENT-DRIVEN AND OPERATIONAL SPILLAGE

20 Trips, Restart and Temporary Product Removal

Material found beside a conveyor does not always result from recurring loss during normal operation. Following a trip, blockage or similar operating event, a conveyor may remain loaded and, in some circumstances, product may need to be removed temporarily from the belt before restart can be achieved. This type of clearing was recorded as applicable at Sample Conveyor 3 during the inspection review. To the best of the inspector's knowledge, product removed for this purpose would normally be expected to be returned to the conveyor or process during subsequent clean-up where practicable. The audit did not review South32's formal trip/restart procedure and therefore does not establish how frequently this type of event occurs or the detailed controls that apply.

This distinction is useful when interpreting material accumulation around the plant. A deposit associated with a documented trip or restart event should not automatically be treated as evidence of recurring mechanical spillage, while repeated accumulation outside such events may still require investigation of the underlying source. The applicable South32 procedure should therefore confirm when temporary product removal is permitted, the required isolation, access and restart controls, authorized responsibilities, temporary handling or placement arrangements, and the approved method for recovering or reintroducing the material into the process. These details should be confirmed from the applicable site procedure rather than inferred from the audit.

21 Housekeeping — Recovery, Access and Ongoing Control

Housekeeping remains an important part of managing material that has already accumulated around the conveyor system. It helps maintain access, recover displaced product and restore affected areas following recurring accumulation or individual operating events. Manual shoveling and wheelbarrow-based removal were reported as methods used in some areas, although these activities were not directly witnessed during the audit. Where practicable, recovered solid product is understood to be returned to the conveyor or process, but the audit did not establish one approved recovery route that applies universally across all areas.

The inspection also indicated that housekeeping practice varies between locations. Some areas appeared to be cleaned relatively regularly, while others appeared to be cleared more on a condition-driven basis as accumulation developed. Water wash-down was reported in some areas, although the receiving drainage or process route was not established for every location. At the Screens, a local collection sump may receive some clean-up water and fines/slurry, but this arrangement should be regarded as specific to that area unless its wider applicability is confirmed.

Housekeeping can therefore manage the accumulated consequence and maintain a workable operating environment, while recurring accumulation can provide useful feedback on whether the underlying material-loss controls are effective. Area-specific responsibility, cleaning frequency or condition-based triggers, approved product-recovery routes and wash-down receiving arrangements should be confirmed through the applicable South32 procedures. A recently cleaned area may show that the consequence has been removed, but it does not by itself establish that the condition responsible for the accumulation has been corrected.

CONTROL PRINCIPLE

Housekeeping addresses the accumulated consequence. Source control aims to reduce recurrence.

PART IX — WHAT THE COMBINED EVIDENCE MEANS

22 Consolidated Engineering Interpretation

Taken together, the findings indicate that the observed spillage and process losses are not explained by one common defect or one common mechanism. Instead, the inspection shows a recurring material-containment theme expressed through several system-wide and location-specific conditions. A common control objective can therefore be applied across the system, while the physical response should remain appropriate to the mechanism, available evidence and local operating geometry at each location.

Table 9 — Consolidated engineering interpretation

Level	Mechanism / condition	Engineering meaning
Systemic / recurring	Fines/carryback; variation in cleaner condition and effectiveness; incomplete retention of cleaner-removed fines at some discharge locations.	The recurring opportunity is to improve cleaner function and retain removed fines within the intended process at applicable discharge locations, reducing reliance on repeated downstream clean-up.
Local / geometry-specific	W10 coarse/mixed-product loss outside the carrying path; Sample Conveyor feed/product-size/loading/belt-position interactions; U4 lateral belt-position/tracking condition with one-sided coarse-product accumulation; L3 alignment/tracking concern with separate fines accumulation.	These locations require assessment of the relevant feed, loading, belt-position and local geometry so that confirmed contributors can be addressed and the appropriate correction or containment arrangement can be finalized. One identical hardware solution should not be assumed.
Water / structural	Active process-water escape at the Screens; wet-fines/slurry conditions at CV01 and CV04 with the initiating water source unresolved; structural deterioration at the Screens and L1.	Water-loss mechanisms and structural condition require location-specific investigation and assessment. Repeated wetting may be relevant to continued exposure, but the available evidence should not be used to assume one common cause for all deterioration.
Operational	Temporary product removal associated with trip/restart clearing; area-specific housekeeping and recovery practices.	Applicable South32 procedures should confirm trip/restart handling, recovery/reintroduction and local housekeeping arrangements, while event-driven deposits remain distinguishable from recurring source-generated loss where practicable.

Across the inspected system, fines accumulation was present throughout the conveyor route, although the extent and immediate condition varied considerably between locations. Carryback is the strongest recurring mechanism, together with variation in cleaner condition, cleaning effectiveness and the containment of material removed at some discharge locations; however, the report does not assume that every individual fines deposit has the same sole origin. Alongside this system-wide fines condition, several local mechanisms influence how and where other material is lost. W10 shows coarse and mixed product outside the intended carrying path within a confined area; the Sample Conveyors illustrate the combined influence of feed, product size, loading, available carrying width and belt position; and U4 shows a relationship between lateral belt position and one-sided coarse-product accumulation. At L3, an alignment/tracking concern and fines accumulation are both present, but the available evidence does not establish the same relationship between them.

Water and structural findings likewise require their own location-specific interpretation. Wet-fines/slurry conditions are present at CV01 and CV04 without the initiating water source being established, while active process-water escape is directly observed at the Screens. The Screens and L1 also contain structural conditions requiring competent assessment, with the available evidence defining the visible deterioration rather than remaining structural capacity or complete causation. Trip/restart product removal and housekeeping form a further operational layer, helping explain and manage some deposits without being treated as one common mechanical spillage mechanism.

The most important systems insight remains that the visible deposit location may be a consequence location rather than the location at which the loss first begins. The combined findings therefore support a common control objective without implying one common physical solution: understand the relevant mechanism and operating condition, address confirmed primary contributors, and keep product and process water within their intended routes using controls suited to the local geometry. This approach forms the basis of the engineering control philosophy developed in the following section.

PART X — THE ENGINEERING CONTROL PHILOSOPHY

23 Engineering Control Philosophy — Keep the Combined Product Inside the Process

CORE PRINCIPLE

The overarching aim is to keep combined product — fines and coarse/lump ore — within the intended material-handling path throughout transit, and to retain process water within the screening and processing systems where it is intended to remain, using controls suited to the specific loss mechanism and local geometry.

23.1 Fines / Carryback

For fines and carryback, the control focus is to maintain the fines within the intended process from one conveyor and transfer stage to the next by improving the cleaning function at relevant discharge locations and keeping material removed by the cleaner within the downstream process route. Because fines accumulation was observed throughout the inspected conveyor ecosystem, the control philosophy is system-wide even though the physical implementation remains location-specific. The following principles apply:

- Remove residual fines at appropriate discharge locations before they progress onto the return run.
- Maintain effective cleaner contact and the required adjustment / wear-compensation function through service.
- Retain cleaner-removed fines within the intended chute / discharge process stream.
- Use the CV09 pilot to demonstrate the required cleaner-and-containment functional standard, then systematically review the remaining relevant conveyor head/discharge locations and progressively adapt the demonstrated function where local geometry, operating duty and maintenance requirements confirm applicability.

23.2 Coarse / Mixed Product

For coarse and mixed product, the first step is to understand the feed, loading, carrying geometry and belt-position relationship before additional containment is selected. The following principles apply:

- Establish the intended duty where available and assess representative feed quantity / variation, product-size distribution and loading position / trajectory relative to the usable carrying geometry.
- Maintain suitable belt centering / tracking and usable carrying margin, and quantify and address confirmed mechanical contributors where required.
- Where lateral product loss or a defined belt-edge containment requirement is established, use belt-compatible side skirting / spill sealing at the carrying edge rather than a remote external barrier. W10 is the preferred demonstration application for this principle. At Sample Conveyors 2 and 3, the same skirting principle should be implemented after the relevant feed/product/loading and belt-position/tracking conditions have been quantified and corrected or adjusted. At other conveyor locations, the principle should be considered and adapted where a comparable containment need and suitable local geometry are confirmed.
- Assess any skirt / seal arrangement under representative operation for belt-compatible contact or clearance, tracking accommodation, product trapping, wear, clean-out and maintenance access.

23.3 Transfer Points and Process Water

Transfer-point and water-related losses should be addressed by keeping material and process water within their intended routes and by establishing the local source mechanism before secondary modifications are defined. The following principles apply:

- Keep transfer material within the intended chute / process path and address confirmed source contributors before adding secondary controls.
- At the Screens, use the planned shutdown investigation to establish the internal water-escape mechanism, correct the confirmed condition and verify containment under representative operation.
- Where wet-fines / slurry conditions are present but the initiating water source remains unresolved, as at CV01 and CV04, assess the local water and material relationship before assigning a specific cause.

23.4 Operational Events

Trips, restart preparation and housekeeping can create or manage deposits that are operationally different from recurring mechanical spillage. These activities should therefore be governed and recorded in a way that keeps temporary handling distinct from recurring source-generated loss. The following principles apply:

- Where product needs to be removed temporarily to facilitate restart after a trip or blockage, keep it under controlled temporary handling and recover or reintroduce it through the approved process route.
- Confirm the applicable South32 procedure, authorized responsibilities, isolation / access / restart controls and approved recovery or reintroduction method.
- Keep event-driven deposits distinguishable from recurring source-generated spillage where practicable, and use housekeeping as consequence management and performance feedback rather than as evidence of source correction.

Across these mechanism families, the overall control sequence can be summarized as: establish the relevant loss mechanism → verify the operating condition or geometry → address confirmed primary contributors → provide the required process routing or primary containment → add residual containment only where needed → recover accumulated consequences safely → confirm representative operating performance and retain an as-left / monitoring record.

PART XI — RECOMMENDED ACTION PROGRAMME

24 Programme Structure and Implementation Principle

The recommended programme follows the control philosophy established in Part X and converts the audit findings into staged work that can be verified, implemented and checked in operation. The sequence begins with demonstration of the cleaner / fines-containment functional standard at CV09 and restoration of basic cleaner function at CV06, then moves through location-specific feed, tracking and containment actions, water and structural assessments, systematic wider review and progressive application of demonstrated controls, and supporting trip / restart and housekeeping governance. Conceptual arrangements remain subject to verified site geometry and the appropriate South32, OEM, responsible-engineer or competent-authority review.

Where this programme requires verification under representative operation, the applicable close-out plan should define the operating conditions against which performance will be assessed before the intervention is closed. These should reflect the operating variables relevant to the specific mechanism, such as normal load/product condition, belt or equipment position and wet/dry condition where applicable. Measurable or observable acceptance criteria and the monitoring period should be agreed by South32 and the responsible maintenance/OEM/design parties before final close-out. Numerical acceptance limits should be taken from the applicable verified design/OEM/site requirements rather than inferred by this audit

24.1 Priority 1A — Demonstrate the Cleaner / Fines-Containment Function at CV09

Engineering objective: develop and trial a practical cleaner-and-containment arrangement that reduces downstream carryback, removes residual fines before the return run and keeps cleaner-removed material within the existing head / discharge-chute product flow, thereby establishing the functional standard for subsequent systematic review of the remaining relevant conveyor head/discharge locations.

Required action sequence

A staged approach is recommended so that the as-found condition, detailed design and operating performance remain directly comparable:

1. Establish the as-found baseline by recording the existing cleaner arrangement and contact condition, head-pulley / chute geometry, access and guarding, downstream carryback, and representative wet / dry operating condition.
2. Develop the detailed design from verified site geometry around the accepted functional concept: pulley-supported cleaning inside the head / discharge chute, two effective cleaning stages where practical, one effective stage where verified constraints prevent two, suitable belt-width coverage, spring-loaded / self-tensioning wear compensation, positive travel limitation and practical maintenance access.
3. Complete OEM / responsible-engineer selection and detailed design of the final cleaner type, materials, contact forces, supports, adjustment arrangement, structural attachments, guarding, dimensions and tolerances, then install and commission the system under the applicable controls. Tru-Trac Rollers (Pty) Ltd may be considered as an indicative supplier for the required primary/secondary belt-cleaner and tensioning functions; however, South32 retains discretion to use any preferred supplier/OEM capable of satisfying the functional and site-specific requirements defined in this report.
4. Monitor representative operation for an agreed period and compare downstream carryback, retention of cleaner-removed material, cleaner contact / wear behavior and maintenance / access condition with the pre-trial baseline. Once the required function is demonstrated, use the monitoring record to begin the systematic applicability review of the remaining relevant conveyor head/discharge locations.

Functional acceptance / close-out criteria

The pilot can be considered ready for close-out when the following outcomes have been demonstrated under representative operation:

- Downstream carryback is materially reduced relative to the pre-trial condition.
- Cleaner-removed wet and dry residual fines remain within the existing head / discharge-chute process flow.
- Stable cleaner contact and cleaning function are maintained through the agreed wear / operating condition without unacceptable belt damage or abnormal cleaner wear.
- Inspection, adjustment and wear-element replacement remain practical, with no unacceptable guarding, material-trapping or access conflict introduced.
- The monitoring record is sufficient to establish the demonstrated functional standard and support systematic review and progressive adaptation at the remaining relevant conveyor head/discharge locations, subject to local geometry and operating requirements.

This recommendation is supported by the CV09 evidence presented in the report, including CV09-REF-002, CV09-REF-003, CV09-REF-005 and the operating-video evidence CV09-REF-006 / REF-008, together with conceptual figures CTD-005 and CTD-006. The audit defines the required function; final cleaner make / model, materials, forces, spring characteristics, supports, adjustment range, travel limits, guarding, structural attachments and fabrication details remain subject to OEM / responsible detailed design.

24.2 Priority 1B — Restore and Verify CV06 Cleaner Function

Engineering objective: restore the CV06 cleaner to a complete, mechanically supported and maintainable operating arrangement, and verify that the required cleaning function is achieved. The housing / support condition should be investigated as a possible contributor without treating it as a visually proven root cause.

The following restoration and verification sequence is recommended:

1. Inspect the complete cleaner assembly and support arrangement, recording the housing, displaced bush / insert, cross-shaft / support, blade coverage and adjustment / retention condition.
2. Restore or replace defective and missing components through the applicable maintenance / OEM process, and review the displacement / blade-loss history where reliable records are available.
3. Verify blade coverage and contact, support integrity, retention and adjustment after restoration, then observe representative operating performance before close-out.

Close-out should demonstrate that the support defect has been addressed, the required blade elements are present across the intended working width, cleaner support / contact / adjustment is functional, and representative operation has been recorded. Any remaining carryback or maintenance concern should be carried forward for action. CV09 remains the preferred pilot location for demonstrating the improved cleaner-and-containment functional standard.

25 Priority 2 — Local Material-Loss and Alignment / Tracking Actions

Priority 2 addresses the location-specific material-loss and alignment / tracking conditions identified in the preceding sections. The appropriate response varies by location and should follow the evidence available at each asset. At L3, for example, the tail-end alignment / tracking concern and the fines accumulation should continue to be assessed separately because the available evidence does not establish a direct coarse-product loss relationship.

25.1 U4 Tripper Car — Tracking / Alignment Verification and Correction

Objective: quantify the U4 belt-position / tracking relationship, address confirmed mechanical contributors where required, and maintain acceptable centred tracking through the normal tripper travel before reassessing the one-sided coarse-product loss and the separate fines condition.

The following verification and correction sequence is recommended:

1. Record and quantify the as-found belt position and relevant tail-pulley / alignment relationship using fixed, repeatable references.
2. Inspect the applicable tail pulley, carrying and return idlers, tension / training arrangement, tripper structural geometry and local interference or product accumulation, and address confirmed mechanical contributors rather than simply repositioning the belt.
3. Verify acceptable centred tracking throughout normal tripper travel under representative loading and operation, retain the as-left reference record, and then reassess the one-sided coarse-product loss and the separate fines condition.

Close-out should demonstrate acceptable centred tracking through the normal tripper travel under representative operation, with the previously observed one-sided coarse-product loss materially reduced or otherwise explained by a remaining verified condition. The separate fines condition and the U4-specific cleaner / discharge arrangement should be reviewed independently. Relevant report evidence includes UGU4-REF-008, REF-009, REF-004 and REF-007 for the tracking / coarse-product relationship, with REF-005 and REF-006 providing the separate fines context.

25.2 Sample Conveyor 3 — Feed, Tracking and Side-Containment Development

Objective: reduce coarse / mixed-product loss by assessing feed quantity and variation, product-size distribution, loading position, usable carrying geometry and the belt-position / tracking condition together before the proposed side-skirt/spill-seal arrangement is finalized.

The following assessment sequence is recommended:

1. Establish the intended / design duty where available and assess the existing feed-control arrangement, representative feed quantity / variation, product-size distribution and loading position / trajectory relative to the narrow carrying geometry.
2. Quantify the visually substantial captured lateral offset using fixed references, inspect alignment, tension / training, interference and entry-zone conditions, and address confirmed primary contributors before verifying centred operation under representative loading.
3. Reassess feed, product-size and loading controls together with the corrected usable carrying geometry, then define and install or adapt the proposed belt-compatible side skirting / spill sealing to suit the verified operating condition and lateral-containment duty.

Close-out should show that the design / feed / product / loading basis has been sufficiently documented, the lateral offset has been quantified, confirmed contributors have been addressed or formally carried forward, and acceptable centred operation has been demonstrated. Coarse / mixed-product loss should be reassessed under representative operation, and the installed/adapted skirt / seal should be belt-compatible and maintainable. Relevant report evidence includes LoadoutStationSample3-REF-002 through REF-007, with REF-006 / REF-007 providing the captured belt-offset views and REF-012 providing wider consequence context.

25.3 Sample Conveyor 2 — Feed / Loading and Belt-Position Verification Before Side-Containment Implementation

Objective: assess how feed quantity and variation, product size, loading position, usable carrying geometry and the captured lateral belt-offset condition combine at Sample Conveyor 2, without assuming a single root cause, and finalize the proposed side-skirt/spill-seal arrangement after confirmed primary contributors have been addressed.

The following assessment sequence is recommended:

1. Establish the intended duty and actual feed-control / metering arrangement, and document representative feed and product-size distribution, loading trajectory / centering and usable belt geometry.
2. Quantify belt position / tracking under representative operation and inspect the congested entry / guide interaction zone, addressing confirmed feed, loading, alignment, tracking or interference contributors where required before verifying centred operation.
3. Reassess the corrected carrying geometry together with the remaining feed and product controls, then define and install the proposed belt-compatible side skirting / spill sealing to suit the verified operating condition and lateral-containment duty, and verify the completed arrangement under representative operation.

Close-out should show that the feed / product / loading / geometry relationship has been documented, the captured offset has been quantified, confirmed contributors have been addressed or formally carried forward, and acceptable centred operation has been demonstrated. Visible product loss should then be reassessed, with the installed skirt / seal maintaining belt-friendly operation, access and maintainability. Relevant report evidence includes LoadoutStationSample2-REF-001, REF-004, REF-005, REF-006 and REF-007.

25.4 W10 — Source-Control Review and Side-Containment Development

Objective: keep coarse / mixed product on the W10 belt by implementing belt-compatible side skirting / spill sealing as the preferred engineered containment addition, while assessing and addressing confirmed loading, tracking and local-geometry contributors so that the final arrangement is developed around the verified operating condition. Reduced accumulation within the restricted tunnel should follow as a consequence of improved material containment rather than being the design objective itself.

The following verification and development sequence is recommended:

1. Record representative belt, trough and support geometry together with belt position / tracking, and assess loading / feed distribution and other relevant source contributors.
2. Address confirmed primary loading, tracking or geometry contributors where practicable and establish the corrected operating condition and lateral-containment duty that the skirting arrangement must accommodate.
3. Develop and install flexible belt-compatible side skirting / spill sealing along the inside of the carrying width, with the lower sealing relationship, extent, support and wear-adjustment arrangement determined from the verified geometry through the selected OEM / responsible design.

Close-out should demonstrate that relevant primary contributors have been investigated and addressed where confirmed, and that the skirt / seal retains product on the belt without unacceptable rubbing, gouging, product trapping, inaccessible build-up, wear or maintenance / access conflict. Representative tunnel accumulation should be reassessed under operation. The conceptual basis is shown in CTD-010 and CTD-011, with W10-REF-002, REF-010, REF-006 and REF-009 providing the report evidence context.

25.5 L3 — Quantitative Alignment / Tracking Verification

Objective: quantify the L3 tail-end alignment / tracking relationship and correct a confirmed defect where required, while continuing to assess the fines accumulation as a separate condition.

The following verification sequence is recommended:

1. Establish fixed references and measure the tail-end belt / pulley / support relationship, inspect applicable idler, alignment, tension / training and interference contributors, and observe representative live tracking.
2. Correct any verified alignment / tracking defect and retain the as-left geometry and operating belt position, then reassess the fines / material accumulation separately without assigning an unsupported coarse-product causal relationship.

Close-out should record the quantified geometry, any confirmed defect and its disposition, representative live tracking and the as-left references. The fines condition should remain separately assessed. Relevant report evidence includes LoadoutStationL3-REF-001 for the alignment / tracking concern and REF-002, REF-004, REF-007 and REF-008 for the separate fines-accumulation context.

26 Priority 3 — Water Containment and Structural Actions

Priority 3 addresses two distinct needs identified in the audit: retaining process water within its intended route at the Screens, and obtaining competent assessment of the structural conditions recorded at the Screens and L1. These actions are related through the operating environment but should be investigated and closed out according to their own evidence and engineering requirements.

26.1 Screens — Identify the Water-Escape Mechanism Before Modification

Objective: retain process water and entrained wet fines within the screening / chute process by using the externally observed escape location to guide a safe shutdown investigation, identifying the internal loss mechanism, correcting the confirmed condition and verifying containment after recommissioning.

The following investigation and verification sequence is recommended:

1. Use Screen-REF-016 and REF-017 to localize and mark the external water-emergence area observed during operation.
2. Isolate the screen and inspect the corresponding internal area during a planned shutdown, identifying the point or mechanism by which water leaves the intended screening / washing / chute process path and recording the relevant geometry and condition.
3. Correct the confirmed water-loss condition, confirm the intended process-water / drainage route, recommission the equipment and re-check the previously identified external area under representative operation.

Close-out should include a documented internal escape mechanism, corrective work that addresses the confirmed condition, and representative operating evidence that water remains within the intended process route at the previously observed location. Repeatedly wetted equipment and steelwork should continue to be assessed separately. Relevant report evidence is Screen-REF-016 and REF-017. Any final chute, drainage or internal modification should be based on the mechanism established during the shutdown inspection rather than assumed in advance.

26.2 Screens — Structural Support Assessment

Objective: establish the structural role, current condition and remaining adequacy of the cut / notched and corroded screen-support members through competent structural assessment, without assuming why the historical alterations were made.

The structural assessment should include the following minimum steps:

1. Appoint a competent structural authority to identify the affected members and load paths, and record member dimensions, cut / notch / opening geometry, corrosion and section loss, connection condition and relevant deformation or deterioration.
2. Review available modification, maintenance and equipment-replacement records where available, determine current adequacy and any required interim controls, and develop and implement the approved repair, strengthening, replacement or reinstatement response.

Close-out should include the competent structural assessment, any available modification history, the defined adequacy / capacity position and required controls, implementation or formal control of the approved intervention, and final as-left inspection with structural acceptance. Detailed plates, welds, member sections, allowable loads and fabrication details remain the responsibility of the competent structural design process. Relevant report evidence includes Screen-REF-009 and REF-010 for the altered support members, together with REF-002, REF-003 and REF-008 for the corrosion / deterioration condition.

26.3 L1 — Structural Condition and Belt-Clearance Verification

Objective: address the documented L1 structural deterioration, verify the previous belt-contact / clearance relationship under representative operation, and reduce recurring product, fines and moisture retention around repaired interfaces while keeping these mechanisms appropriately separated.

The following structural and mechanical verification sequence is recommended:

1. Complete a competent structural assessment of the affected concrete, exposed reinforcement, steel members, supports and connections, recording concrete loss, steel section loss or perforation and the inputs required for repair design, together with any necessary interim controls.
2. Record the wear-slot / contact geometry and relevant belt-to-structure clearance, and observe representative belt tracking and clearance. Address a current mechanical contributor where it is demonstrated; otherwise retain the qualification that the wear represents previous or intermittent contact.
3. Implement the approved structural repair or rehabilitation and formal as-left acceptance, reduce recurring product / fines / moisture accumulation around repaired interfaces, and include the local condition in follow-up monitoring.

Close-out should confirm that structural adequacy has been assessed and the approved disposition implemented or formally controlled, representative belt clearance / tracking is acceptable, and no unsupported claim of continuous present contact has been introduced. Recurring accumulation and moisture retention should be addressed as a separate environmental control. Relevant report evidence includes LoadoutStationL1-REF-004 and REF-005 for concrete deterioration, REF-008 and REF-009 for steel deterioration, REF-007 for previous / intermittent contact, and REF-013 and REF-017 for the accumulation environment.

27 Priority 4 — Extend Demonstrated Controls Where Applicable

Following successful CV09 pilot verification, the relevant conveyor head/discharge locations across the inspected material-handling route should be systematically reviewed against the demonstrated cleaner-and-containment functional standard. Where the local carryback condition, geometry, operating duty, material behavior and maintenance/access requirements confirm applicability, the function should be progressively adapted and implemented. This does not require one identical cleaner installation at every conveyor; the system standard is the required cleaning-and-containment function, while the hardware and installation arrangement remain location-specific. U4 should continue to be treated as a geometry-specific exception unless a later detailed review demonstrates an appropriate equivalent arrangement. Separately, once the W10 side-skirt/spill-seal principle has demonstrated satisfactory representative performance, other underground and surface conveyor locations should be reviewed for comparable lateral-product-loss or belt-edge containment requirements. Where the local mechanism and geometry confirm suitability, the demonstrated skirting principle should be adapted and applied. Potential applications include, but are not limited to, restricted or confined underground conveyor sections and the Sample Conveyors following completion of their prerequisite feed/product/loading/tracking corrections.

The wider review should therefore distinguish between a system-wide fines-control standard and location-specific side containment. The following checks apply:

- For cleaner / fines-control applications, review each relevant head/discharge location against the demonstrated functional standard, including the available pulley-supported cleaning position, practical stage count, blade coverage and contact, wear compensation / self-tensioning, positive travel limitation, component retention, containment of removed material, maintainability and actual downstream carryback. Where applicability is confirmed, develop the site-specific implementation rather than leaving the condition as an optional future consideration.
- For side-containment applications, use inspection and operating evidence to identify locations with comparable lateral product loss or a defined belt-edge containment requirement. Confirm the local material-loss mechanism, feed and loading condition, belt position / tracking, carrying geometry, required side-containment duty, belt-compatible contact or clearance, clean-out and maintenance access before adapting a skirt / seal arrangement.

28 Priority 5 — Operational and Housekeeping Governance

Operational and housekeeping controls remain important for event-driven deposits, safe access, material recovery and performance feedback. They support the engineering programme but do not replace source correction where a recurring mechanism can be addressed. Because the audit did not review the mine's complete formal procedures, this section identifies the matters that should be confirmed by South32 rather than prescribing site-specific roles, thresholds, fixed frequencies or universal recovery routes.

28.1 Trip / Restart Product Handling

The applicable written trip / restart / belt-clearing procedure should confirm when temporary product removal is authorized, the required isolation, access and restart controls, authorized responsibilities, temporary handling or placement arrangements, the approved recovery / reintroduction method, and how event-driven deposits are distinguished from recurring source-generated spillage. The source-recorded understanding is that temporary off-loading may be used where retained belt load hinders restart, with removed product expected to be returned to the conveyor or process during clean-up where practicable.

Close-out should confirm the approved procedure and responsibilities, the temporary handling and recovery route, and a practical means of distinguishing event-driven deposits from recurring spillage in operating history where practicable.

28.2 Housekeeping / Recovery

Area-specific housekeeping responsibility, cleaning frequency or condition-based trigger, approved solid-product recovery / reintroduction route and, where wash-down is used, the receiving drainage or process route should be confirmed through the applicable South32 procedures. Manual shoveling and wheelbarrow-based removal were reported but not directly witnessed. The Screens building sump may form part of that area's clean-up-water / fines recovery subject to confirmation, but it should not be assumed to represent a plant-wide recovery system. Repeat-accumulation records can provide useful feedback on whether source-control measures are working.

Governing principle: Housekeeping addresses the accumulated consequence; source control aims to reduce recurrence. A recently cleaned area may show that the consequence has been removed, but it does not by itself establish that the initiating condition has been corrected.

28.3 Programme Close-Out Rule

Each material action should be closed out through a traceable sequence that links the as-found condition → verification or measurement → corrective design or control → implementation → representative operating verification → as-left record → monitoring and closure. Maintenance sign-off, installation completion or a cleaned area should be treated as an intermediate milestone unless the intended functional outcome has also been demonstrated.

28.4 Consolidated Priority Action Register

The following register summarizes the principal actions and the verification or design basis that should accompany each priority:

Table 10 — Consolidated priority action register

Priority	Action	Type	Verification / design requirement	Primary evidence / case
1A	CV09 cleaner / fines-containment pilot	Design development / pilot	Record as-found geometry and carryback; develop pulley-supported cleaner arrangement inside head / discharge chute; prefer two effective stages where practical; provide wear compensation / self-tensioning and travel limitation; verify retention of removed fines and representative performance.	CV09
1B	Restore and verify CV06 cleaner function	Maintenance / operating verification	Restore missing / defective cleaner components; verify support, retention, blade coverage, contact and adjustment; observe representative operation.	CV06
2	U4 belt-position / tracking verification and correction	Corrective maintenance + verification	Quantify as-found relationship; inspect and address confirmed contributors; verify centred tracking through normal tripper travel; reassess one-sided coarse-product loss and separate fines condition.	U4
2	Sample Conveyor 3 feed / product / loading / belt-position verification	Process / mechanical verification	Establish design/feed/product/loading basis; quantify captured REF-006/REF-007 offset; address confirmed contributors and verify centred operation; then define/install the proposed belt-edge skirting to suit the corrected operating condition and verify representative performance.	Sample 3

Priority	Action	Type	Verification / design requirement	Primary evidence / case
2	Sample Conveyor 2 feed / loading / belt-position verification	Process verification	Establish duty and feed-control basis; quantify captured REF-004/REF-005 offset; inspect entry/guide congestion; address confirmed contributors and verify centred operation; then define/install the proposed belt-edge skirting to suit the corrected operating condition and verify representative performance.	Sample 2
2	W10 source review + belt-edge side containment	Engineering verification / concept development	Assess loading, tracking and local geometry; correct confirmed primary contributors; develop and install the preferred belt-edge side-skirt/spill-seal arrangement around the verified operating condition; verify belt compatibility, access, maintainability and representative containment performance.	W10
2	L3 alignment / tracking verification	Mechanical verification	Quantify tail-end geometry and live tracking; correct confirmed defect if present; retain as-left references; reassess fines accumulation separately.	L3
3	Screens water-escape investigation	Shutdown inspection / design input	Use REF-016 / REF-017 to localize external escape; inspect corresponding internal area during shutdown; correct confirmed mechanism; verify containment after recommissioning.	Screens
3	Screens structural assessment	Structural	Identify member roles and load paths; record cuts / notches, corrosion, section loss and connections; complete competent adequacy and repair disposition; obtain formal as-left acceptance.	Screens
3	L1 structural + belt-clearance verification	Structural / mechanical	Assess concrete and steel condition; record wear / contact geometry; verify representative belt clearance / tracking; implement approved repair; reduce recurring accumulation and moisture retention.	L1
4	Systematic applicability review and progressive rollout of demonstrated controls	Engineering review / progressive implementation	After satisfactory CV09 pilot performance, systematically review relevant conveyor head/discharge locations against the demonstrated cleaner-and-containment functional standard and adapt the function where local geometry, duty and maintenance requirements confirm applicability. Apply the demonstrated W10 side-containment principle where inspection or operating evidence identifies a comparable lateral-product-loss or belt-edge containment requirement and the local mechanism and geometry confirm suitability.	System-wide relevant head/discharge locations + other applicable side-containment locations
5	Trip / restart product-handling governance	Operational	Confirm formal procedure, authorization, isolation / access / restart controls, temporary handling and approved recovery / reintroduction route.	System-wide
5	Housekeeping / recovery governance	Operational	Confirm area responsibility, cleaning frequency or condition trigger, approved recovery route and wash-down receiving route where used; retain repeat-accumulation feedback.	System-wide / area-specific

PART XII — CONCLUSION

29 Conclusion

The Combined Product Spillage Audit indicates that the visible spillage and process losses recorded across the inspected Wessels Mine material-handling route are better understood as a material- and process-containment issue expressed through several different mechanisms, rather than as the result of one common defect. Fines spillage was observed throughout the inspected conveyor ecosystem, with the extent varying considerably between locations and no inspected conveyor section completely free of fines accumulation. The presence of fines was therefore system-wide, while the severity, local condition and immediate mechanism were not necessarily the same at every location. Carryback remains the strongest recurring mechanism identified across the system, but the report does not assume that every individual fines deposit arose from the same sole cause.

For recurring fines and carryback, the findings support a system-level improvement in both the cleaning function and the retention of cleaner-removed material within the intended process. At CV06, the immediate priority is to restore and verify the installed cleaner function. CV09 provides the preferred pilot for demonstrating pulley-supported cleaning within the head/discharge-chute containment, with wear compensation, travel control and maintenance access developed through detailed design. CV09 also carries fines separated during the washing/screening process to the downstream fines collection/handling point, making it a useful reference for the wider end-to-end objective of keeping fines within the process until they reach the point at which they are intentionally collected and handled. Once the CV09 pilot has demonstrated the required cleaning-and-containment function under representative operation, the relevant conveyor head/discharge locations across the inspected route should be systematically reviewed and progressively brought to the same functional standard where local conditions confirm applicability. The intention is to standardize the function — effective cleaning, reliable contact through service and retention of removed fines within the downstream process route — rather than install identical hardware at every conveyor.

Coarse and mixed-product loss requires a different emphasis. At W10, the demonstrated lateral product loss and restricted tunnel environment support belt-edge side skirting or spill sealing as the preferred engineered containment addition. Relevant loading, tracking and local geometry should still be verified and confirmed contributors corrected so that the final skirting arrangement is developed around the verified operating condition. At Sample Conveyors 2 and 3, the same skirting principle is proposed after feed quantity/variation, product-size distribution, loading position and belt-position/tracking or alignment have been quantified and corrected or adjusted where required; the corrected condition should determine the final skirting extent and duty. Once the W10 principle has demonstrated satisfactory representative performance, it should also be considered and adapted at other underground or surface conveyor locations where a comparable lateral-containment need is identified and the local mechanism and geometry confirm suitability, including but not limited to confined or restricted underground conveyor sections. U4 supports a relationship between lateral belt position and one-sided coarse-product accumulation, although the complete mechanical reason for the tracking condition still requires quantitative verification. At L3, the tail-end alignment/tracking concern and the separate fines accumulation should continue to be assessed independently.

Water-related and structural findings also require location-specific follow-up. At the Screens, active process-water escape was observed during operation, but the corresponding internal source or defect remains to be established during a planned shutdown before the final corrective modification is defined. Wet-fines/slurry conditions are also present at CV01 and CV04, although the initiating water source is unresolved. Structural deterioration at the Screens and L1 requires competent assessment because photographs can establish visible condition but not remaining structural capacity or complete causation. At L1, the wear evidence is consistent with previous or intermittent belt-to-structure contact while the belt was running clear in the inspected operating condition, so the structural and mechanical questions should remain separated.

The audit further distinguishes recurring source-generated spillage from material that may be removed temporarily during a trip, blockage or restart intervention. These event-driven deposits still require controlled handling and an approved recovery or reintroduction route, but they should not automatically be treated as evidence of recurring mechanical loss at the deposit location. Housekeeping remains important for access and recovery, while a recently cleaned area should be regarded as evidence that the accumulated consequence has been removed rather than, by itself, confirmation that the initiating condition has been corrected.

Across these mechanisms, the common engineering objective is to keep combined product within its intended material-handling path and process water within its intended process route, using controls suited to the actual mechanism and local operating geometry. For fines, this becomes an end-to-end system requirement: residual fines should be removed before entering the return run and cleaner-removed material should remain within the intended downstream process flow through successive conveyor and transfer stages until it reaches the designated collection/handling point. The report therefore does not propose one identical cleaner, skirt or mechanical modification throughout the system. What can be applied consistently is the required function and control philosophy: establish the relevant mechanism and operating condition, address confirmed primary contributors, provide the required source or process containment, add residual containment only where it remains necessary, and confirm the result under representative operation.

The recommended programme combines corrective work with the measurement, detailed design, competent assessment and operating verification needed to turn the report-level concepts into site-ready actions. Matters that remain open — including exact site geometry, quantified tracking conditions, structural capacity, the internal Screens water-loss mechanism, final OEM component detail, applicable operating procedures and representative as-left performance — should be carried forward as verification, design or governance requirements rather than resolved through unsupported assumptions. The intended outcome is a material-handling system in which fines and coarse product remain within their intended routes as far as practicable, the demonstrated cleaner-and-containment function is progressively applied across relevant conveyor discharge/head locations, location-specific residual containment is added where evidence shows it is required, temporary interventions remain controlled and recoverable, and the effectiveness of each intervention is demonstrated and sustained in operation.

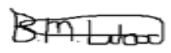
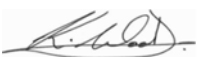
FINAL TAKEAWAY

The long-term objective is to keep product and process water within their intended routes, establish the CV09 cleaner-and-containment function as the reference standard for progressive application at relevant conveyor head/discharge locations, address recurring loss mechanisms at source where they can be established, and use representative operating performance — rather than a temporarily clean area — as the measure of whether the control is working.

30 Technical Review and Report Issue Sign-off

This report represents the controlled culmination of the inspection, evidence review and engineering assessment described herein. The findings, engineering interpretations, recommendations and associated evidence have been reviewed for technical consistency and are considered suitable for controlled issue within the stated scope and limitations of the assessment. Items identified in the report for further measurement, verification, OEM/responsible-design development, competent structural assessment or representative operating confirmation remain implementation and close-out requirements and do not prevent issue of the present assessment. Approval for issue confirms acceptance of the report as an engineering inspection and assessment record; it does not constitute approval of subsequent detailed designs, modifications or site implementation work, which remain subject to the applicable South32, OEM and responsible-engineering review and authorization processes.

Table 11 — Technical review and issue sign-off

Role	Name	Title / discipline	Review status	Date	Signature
Prepared by	Brendan Wood	Projects / Systems	Report preparation	31/08/2026	
Technical reviewer	Kevin Wood	Managing Engineer	Review Complete.	01/09/2026	
Approved for issue	Kevin Wood	Managing Engineer	Approval	01/09/2026	

APPENDICES — TECHNICAL EVIDENCE BACKBONE

Appendix A — Full Asset-by-Asset Audit Matrix

Appendix A provides the report-facing 23-group asset audit matrix. It summarizes each controlled inspection group, the strongest supported condition/mechanism, representative evidence references and the action direction. The historical 197-row EVIDENCE_STORY_MATRIX_V0.1.csv remains part of the detailed development audit trail. Current item-level evidence use is governed by evidence_index_CURRENT.md, the applicable active evidence overlays and the Sample Conveyor 3 REF-013 withdrawal control.

Table 12 — Full asset-by-asset audit matrix

Area	Asset / group	Verified condition / mechanism	Key evidence	Action direction
Underground – Central Block	Underground Crusher	Mixed coarse/lump and fines accumulation; local standing water visible, source unresolved.	UGCrusher-REF-001/-002	Investigate local material/water contributors; maintain source-control/carryback review.
Underground – Central Block	UC05-UC01 continuous system	Distributed dry fines with mixed coarse fragments; severe local mounds and clear comparator sections.	UGUC05_04_03_02_01-REF-005/-006/-007/-015	Treat as systemic recurrence/carryback story; target discharge/cleaner source control and trending.
Underground – Secondary Plant	U3 Conveyor	Local conveyor-adjacent fines/mixed accumulation; roadway coarse rock not automatically conveyor spillage.	UGU3-REF-004/-007	Maintain origin qualification; local inspection/source control as required.
Underground – Secondary Plant	U4 Tripper Car	Visually established lateral mistracking/offset; inspector-recorded tail-pulley/alignment concern unmeasured; one-sided coarse-product consequence plus separate fines condition.	UGU4-REF-008/-009/-004/-007; fines -005/-006; video -010 supporting	Quantify/correct alignment/tracking and verify full normal tripper travel under representative loading; reassess coarse loss and fines separately; cleaner exception reviewed locally.
Underground – Silos / Feeders	U5A Conveyor	Extensive/widespread dry fines; return-side geometry strongly consistent with carryback.	UG5A-REF-002/-004/-006	Priorities cleaner/discharge source-control applicability review.
Underground – Silos / Feeders	U5B Conveyor	Trace-to-localized dry fines overall with some sizeable local deposits.	UG5B-REF-001/-003/-006	Maintain localized classification; inspect source/cleaner only as evidence warrants.
Surface – Screening / Transfer	W10	Substantial coarse/mixed product in restricted tunnel; exact initiating cause unresolved.	W10-REF-002/-010; supporting -006/-009	Verify loading/tracking/geometry and correct confirmed contributors; develop and install the preferred belt-compatible side skirting/spill sealing inside the carrying width around the verified operating condition; verify representative performance.
Surface – Screening / Transfer	CV01	Wet-fines/slurry and pooled water; severe visible roller surface damage; seized state not dynamically proven.	CV01-REF-001/-003	Correct maintenance condition; investigate water/material source without overclaim.
Surface – Screening / Transfer	CV02	Dry fines plus mixed/lump accumulation around transfer/return-side structure.	CV02-REF-001/-003	Supporting fines/carryback/source-control review.
Surface – Screening / Transfer	CV03	No significant spillage in inspected condition; clean comparator.	CV03-REF-002	Retain as time-specific comparator; no escalation.
Surface – Screening / Transfer	CV04	Saturated wet-fines/slurry and multiple piles; extensive/widespread.	CV04-REF-002/-005	Investigate water/material source and containment; source not inferred from stills.
Surface – Screening / Transfer	CV06	Displaced cleaner support bush/insert; scraper blades absent from working width.	CV06-REF-004/-005	Restore and verify cleaner function; investigate history where records permit.
Surface – Screening / Transfer	CV09	Wet fines/slurry, cleaner outside/open to effective head-chute containment, active fallout and residual downstream carryback; preferred pilot for integrated cleaner/fines containment.	CV09-REF-002/-003; video -006	Pilot pulley-supported cleaning inside head/discharge chute; preferred two stages where feasible; self-tensioning/wear stop; representative monitoring; OEM detail design.
Surface – Screening / Transfer	CV14	Trace/localized dry fines/carryback deposits.	CV14-REF-002	Supporting low-severity recurrence; no escalation.
Surface – Screening / Transfer	CV15	Extensive distributed fines/carryback around chute E-3181; stills mainly dry/dried.	CV15-REF-003/-004/-006	Assess cleaner/discharge functionality; retain wet-condition qualification.

Area	Asset / group	Verified condition / mechanism	Key evidence	Action direction
Surface – Screening / Transfer	Screens / screen chutes	Active process-water escape verified; severe corrosion and cut/notched supports verified; exact mechanism/capacity unresolved.	Screen-REF-016/-017 water; corrosion -002/-003 (support -004, extent -008); structural -009/-010 (support -006)	Localize external water emergence, planned shutdown internal-source verification/correction, operating verification; separate competent structural assessment and formal acceptance.
Surface – Loading Station	L1 Conveyor	Extensive spillage; severe concrete/steel deterioration; previous/intermittent belt-contact evidence.	Concrete -004/-005; steel -008/-009; historical contact -007; environment -013/-017	Competent structural assessment/repair acceptance plus representative belt-clearance/tracking verification; reduce recurring accumulation/moisture. Latest running belt clear/no visible mistracking.
Surface – Loading Station	L2 Conveyor	No significant spillage in inspected condition; photo/video comparator.	LoadoutStationL2-REF-004.jpg / REF-005 media	Retain time-specific clean comparator; no chronic claim.
Surface – Loading Station	L3 Conveyor	Tail-end offset supported but unmeasured; fines visible; significant solid-product causal chain not established.	Alignment REF-001; fines REF-002/-004/-007; supporting -008	Quantify alignment/live tracking; correct/disposition without overclaiming product loss.
Surface – Loading Station	L4 Conveyor	Localized dry fines and coarse/lump accumulation; operating stills generally contained.	LoadoutStationL4-REF-003/-004	Local source/containment review as required; no active-fall claim.
Surface – Loading Station	Sample Conveyor 1	Extensive/widespread mixed fines/coarse accumulation; adhered/dried deposits indicate sticky/moist history.	LoadoutStationSample1-REF-011/-008/-005	Supporting system consequence/source-control review; active wetness not proven.
Surface – Loading Station	Sample Conveyor 2	Coarse/lump product occupies substantial narrow carrying geometry; containment loss; captured lateral offset supported by REF-004/005 but magnitude/persistence/root cause unverified.	LoadoutStationSample2-REF-004/-005/-001/-007/-006	Verify design/feed/product/loading basis; quantify/correct tracking/interference; verify centred operation; then define and install/adapt belt-compatible side skirting to suit the verified condition
Surface – Loading Station	Sample Conveyor 3	Substantial coarse/lump spillage and narrow carrying margin; visually substantial lateral offset supported by REF-006/007, unmeasured; root contributor unresolved.	LoadoutStationSample3-REF-002/-003/-004/-005/-006/-007/-012; REF-013 withdrawn	Verify design/feed/product/loading basis; quantify/correct tracking/alignment contributors; verify centred operation; then define and install/adapt belt-compatible side skirting to suit the verified condition. Do not use REF-013.

Appendix B — Detailed Photographic Evidence Register

Appendix B provides the controlled photographic register by asset group. The current active population contains 188 photographs and 8 videos (196 inspection media). LoadoutStationSample3-REF-013 is withdrawn from active use and is therefore not included in the 188-photo count. Current E/S/A placement priorities are governed by evidence_index_CURRENT.md and the applicable Sample Conveyor, U4/L3, Screens and L1 overlays. Repository originals remain master evidence.

Table 13 — Detailed photographic evidence registers

Asset / group	Photos	Canonical range / naming	Principal body candidates	Control note
Underground Crusher	5	UGCrusher-REF-001..005	UGCrusher-REF-001/-002	Repository originals retained. Current body/appendix priority is governed by evidence_index_CURRENT.md and the applicable active overlay; use the smallest sufficient evidence set and preserve item-specific qualifications.
UC05-UC01 continuous system	16	UGUC05_04_03_02_01-REF-001..016	UGUC05_04_03_02_01-REF-005/-006/-007/-015	Repository originals retained. Current body/appendix priority is governed by evidence_index_CURRENT.md and the applicable active overlay; use the smallest sufficient evidence set and preserve item-specific qualifications.
U3 Conveyor	8	UGU3-REF-001..008	UGU3-REF-004/-007	Repository originals retained. Current body/appendix priority is governed by evidence_index_CURRENT.md and the applicable active overlay; use the smallest sufficient evidence set and preserve item-specific qualifications.
U4 Tripper Car	11	UGU4 photos: REF-001..009, -011..012	UGU4-REF-008/-009/-004/-007; fines -005/-006; video -010 supporting	Repository originals retained. Current body/appendix priority is governed by evidence_index_CURRENT.md and the applicable active overlay; use the smallest sufficient evidence set and preserve item-specific qualifications.
U5A Conveyor	9	UG5A-REF-001..009	UG5A-REF-002/-004/-006	Repository originals retained. Current body/appendix priority is governed by evidence_index_CURRENT.md and the applicable active overlay; use the smallest sufficient evidence set and preserve item-specific qualifications.
U5B Conveyor	6	UG5B-REF-001..006	UG5B-REF-001/-003/-006	Repository originals retained. Current body/appendix priority is governed by evidence_index_CURRENT.md and the applicable active overlay; use the smallest sufficient evidence set and preserve item-specific qualifications.
W10	12	W10-REF-001..012	W10-REF-002/-010; supporting -006/-009	Repository originals retained. Current body/appendix priority is governed by evidence_index_CURRENT.md and the applicable active overlay; use the smallest sufficient evidence set and preserve item-specific qualifications.
CV01	4	CV01-REF-001..004	CV01-REF-001/-003	Repository originals retained. Current body/appendix priority is governed by evidence_index_CURRENT.md and the applicable active overlay; use the smallest sufficient evidence set and preserve item-specific qualifications.
CV02	4	CV02-REF-001..004	CV02-REF-001/-003	Repository originals retained. Current body/appendix priority is governed by evidence_index_CURRENT.md and the applicable active overlay; use the smallest sufficient evidence set and preserve item-specific qualifications.
CV03	3	CV03-REF-001..003	CV03-REF-002	Repository originals retained. Current body/appendix priority is governed by evidence_index_CURRENT.md and the applicable active overlay; use the smallest sufficient evidence set and preserve item-specific qualifications.
CV04	5	CV04-REF-001..005	CV04-REF-002/-005	Repository originals retained. Current body/appendix priority is governed by evidence_index_CURRENT.md and the applicable active overlay; use the smallest sufficient evidence set and preserve item-specific qualifications.

Asset / group	Photos	Canonical range / naming	Principal body candidates	Control note
CV06	5	CV06-REF-001..005	CV06-REF-004/-005	Repository originals retained. Current body/appendix priority is governed by evidence_index_CURRENT.md and the applicable active overlay; use the smallest sufficient evidence set and preserve item-specific qualifications.
CV09	5	CV09-REF-001..005	CV09-REF-002/-003; video -006; -008 supplementary	Repository originals retained. Current body/appendix priority is governed by evidence_index_CURRENT.md and the applicable active overlay; use the smallest sufficient evidence set and preserve item-specific qualifications.
CV14	2	CV14-REF-001..002	CV14-REF-002	Repository originals retained. Current body/appendix priority is governed by evidence_index_CURRENT.md and the applicable active overlay; use the smallest sufficient evidence set and preserve item-specific qualifications.
CV15	8	CV15-REF-001..008	CV15-REF-003/-004/-006	Repository originals retained. Current body/appendix priority is governed by evidence_index_CURRENT.md and the applicable active overlay; use the smallest sufficient evidence set and preserve item-specific qualifications.
Screens / screen chutes	14	Screen-REF-001..014	Water -016/-017; corrosion -002/-003 (support -004; extent -008); structural -009/-010 (support -006)	Repository originals retained. Current body/appendix priority is governed by evidence_index_CURRENT.md and the applicable active overlay; use the smallest sufficient evidence set and preserve item-specific qualifications.
L1 Conveyor	18	LoadoutStationL1-REF-001..018	Concrete -004/-005; steel -008/-009; historical contact -007; environment -013/-017	Repository originals retained. Current body/appendix priority is governed by evidence_index_CURRENT.md and the applicable active overlay; use the smallest sufficient evidence set and preserve item-specific qualifications.
L2 Conveyor	6	LoadoutStationL2-REF-001..006.jpg	LoadoutStationL2-REF-004.jpg / REF-005 media	Repository originals retained. Current body/appendix priority is governed by evidence_index_CURRENT.md and the applicable active overlay; use the smallest sufficient evidence set and preserve item-specific qualifications.
L3 Conveyor	8	LoadoutStationL3-REF-001..008	Alignment -001; fines -002/-004/-007; supporting -008	Repository originals retained. Current body/appendix priority is governed by evidence_index_CURRENT.md and the applicable active overlay; use the smallest sufficient evidence set and preserve item-specific qualifications.
L4 Conveyor	8	LoadoutStationL4-REF-001..008	LoadoutStationL4-REF-003/-004	Repository originals retained. Current body/appendix priority is governed by evidence_index_CURRENT.md and the applicable active overlay; use the smallest sufficient evidence set and preserve item-specific qualifications.
Sample Conveyor 1	11	LoadoutStationSample1-REF-001..011	LoadoutStationSample1-REF-011/-008/-005	Repository originals retained. Current body/appendix priority is governed by evidence_index_CURRENT.md and the applicable active overlay; use the smallest sufficient evidence set and preserve item-specific qualifications.
Sample Conveyor 2	8	LoadoutStationSample2-REF-001..008	REF-004 + REF-001 opening; REF-004 + REF-005 offset/product-size; REF-007 loading; REF-006 congestion	Repository originals retained. Current body/appendix priority is governed by evidence_index_CURRENT.md and the applicable active overlay; use the smallest sufficient evidence set and preserve item-specific qualifications.
Sample Conveyor 3	12 active	LoadoutStationSample3-REF-001..012 active; REF-013 withdrawn	REF-002/-003/-004/-005/-006/-007/-012; principal offset -006/-007	REF-013 withdrawn from active population because it belongs to another conveyor/belt; do not use. Current priorities governed by Sample Conveyor overlay/withdrawal control.

Appendix C — Video Evidence Register

Appendix C lists the 8 controlled videos individually. Video is used where dynamic operating state materially strengthens the evidence; extracted static frames remain derivatives of the original video and retain the original canonical evidence ID. Reader-facing access is provided in the report body for CV09-REF-006, CV09-REF-008, Screen-REF-016 and Screen-REF-017 through a visible View Video hyperlink and corresponding QR code. The remaining controlled videos are retained in this register as evidence without body-linked reader access.

Table 14 — Video evidence registers

Asset	Canonical ID	Repository filename	Evidence role	Qualification
CV09	CV09-REF-006	Cv09-Ref-006.mp4	Active wet-material fallout during operation; principal dynamic operating evidence.	Does not define final cleaner/chute geometry.
CV09	CV09-REF-007	Cv09-Ref-007.mp4	Supplementary operating context for wet-fines / cleaner region.	Supporting dynamic evidence; use only where it adds unique context.
CV09	CV09-REF-008	Cv09-Ref-008.mp4	Dynamic cleaner/chute contamination and containment-boundary context.	Supports pilot context; not final design geometry.
Screens	Screen-REF-015	Screen-REF-015.mp4	Supplementary screen operating / water context.	Supporting only unless unique mechanism detail is required.
Screens	Screen-REF-016	Screen-REF-016.mp4	Direct operating evidence of active process-water escape/spray.	Proves active escape, not exact equipment-level cause.
Screens	Screen-REF-017	Screen-REF-017.mp4	Strongest dynamic evidence for localizing the external point of water emergence/escape and showing repeated wetting.	Localizes external emergence; does not identify the internal defect/mechanism or prove sole corrosion causation.
L2	LoadoutStationL2-REF-005 (video)	LoadoutStationL2-REF-005.mp4	Dynamic clean-condition / no-significant-spillage comparator for inspected state.	Duplicate numeric ID exists as JPG; always state media type/exact filename.
U4	UGU4-REF-010	UGU4-REF-010.MP4	Dynamic tail-pulley/belt alignment/tracking context.	Supports inspector-recorded concern; not quantitative geometry.

Appendix D — Measurements / Verification Required

Class	Definition	Application in this report
M	Mechanical / physical-condition verification	Inspection, measurement or recording of physical condition, geometry, alignment, tracking, component configuration, clearances or other mechanical relationships required to establish the as-found/as-left condition.
D	Design / detailed-design development	Verification or information required to define, select or complete the final engineering arrangement through the OEM/responsible-design process.
O	Operating verification	Observation or verification under representative operating conditions to establish actual equipment/material behaviour and confirm the as-left operating condition.
P	Performance / pilot verification	Before/after performance assessment, monitoring or pilot evidence used to demonstrate whether the intended functional outcome has been achieved.
S	Structural assessment	Competent structural investigation, condition/capacity assessment, repair or strengthening disposition and structural acceptance.
C	Control / governance verification	Confirmation of procedures, responsibilities, authorisations, operating controls, recovery arrangements or other governance requirements.
A	Applicability / wider-rollout assessment	Review of a demonstrated control or pilot result to determine whether and how the principle should be adapted at other applicable locations.

Gate	Definition	Required outcome
G1	As-found / diagnostic / design-input gate	Establish the existing condition, relevant mechanism, measurements and design inputs before the final corrective arrangement is defined.
G2	Corrective design / implementation gate	Define, approve and implement the required correction, repair, engineering design or procedural control based on the verified G1 condition.
G3	Operating verification / close-out gate	Demonstrate the as-left condition and intended functional performance under representative operation, with the required records and acceptance before close-out.
G4	Wider-application / rollout gate	Use the demonstrated and documented result to determine whether the control principle is suitable for systematic review, adaptation or wider implementation at other locations.

Table 15 — Measurements / verification required

ID	Case / verification	Class	Gate	Required evidence	Decision / close-out dependency
VER-001	CV09 as-found geometry / controlled pilot design inputs	M/D/O	G1	Head-pulley/chute/access survey; stage/location feasibility; support/slot/tensioning/travel-stop inputs; pre-trial carryback baseline	Before final cleaner/chute design
VER-002	CV09 pilot functional performance	O/P	G3/G4	Representative before/after carryback + retained-discharge evidence + cleaner contact/wear/maintenance record	Before wider rollout
VER-003	CV06 cleaner restoration	M/O	G2/G3	Maintenance repair record + post-repair configuration + operating verification	Before cleaner close-out
VER-004	U4 alignment/tracking/full travel	M/O	G1/G2/G3	As-found/as-left survey + full-travel operating record + loss comparison	Before final tracking close-out
VER-005	Sample 3 design/feed/product/loading/tracking	M/O/D	G1/G2/G3	Design basis; feed/product/loading; usable geometry; REF-006/007 offset/alignment; post-correction centred-operation record; final skirt/seal design inputs and representative containment verification	Before final skirt/seal design and close-out
VER-006	Sample 2 design/feed/product/loading/tracking	M/O/D	G1/G2/G3	Design basis; feed/product/loading; usable geometry; quantify REF-004/005 offset; congestion/interference check; post-correction centred-operation record; final skirt/seal design inputs and representative containment verification	Before final skirt/seal design and close-out
VER-007	W10 source/tracking verification + preferred side-skirt design inputs	M/O/D	G1/G2/G3	Loading/tracking/geometry review; corrected operating condition; required skirt/seal extent and containment duty; OEM skirt/seal detail; representative belt-friendly containment/access/wear verification	Before skirt/seal design / final close-out
VER-008	L3 quantitative alignment/live tracking	M/O	G2/G3	As-found/as-left geometry + live tracking + local post-correction condition	Before alignment close-out
VER-009	Screens water-escape mechanism	O/D/M	G1/G2/G3	REF-016/017 external localization + planned shutdown internal-source/defect record + corrective geometry + post-modification operating verification	Before detailed modification/close-out
VER-010	Screens structural condition/capacity	S/M	G1/G2/G3	Competent assessment: member role/load path, cuts/notches, section loss, connections, remaining adequacy, interim control/repair disposition and as-left acceptance	Before structural acceptance

ID	Case / verification	Class	Gate	Required evidence	Decision / close-out dependency
VER-011	L1 structural + belt clearance	S/M/O	G1/G2/G3	Competent structural record + concrete/steel deterioration + wear/contact geometry + representative belt tracking/clearance + as-left acceptance	Before structural/mechanical close-out
VER-012	Trip/restart governance	C	G2/G3	Approved procedure + authorization/responsibility + isolation/restart controls + controlled temporary handling + approved recovery/reintroduction route	Before operational close-out
VER-013	Housekeeping / recovery governance	C	G3	Area responsibility + scheduled frequency/condition trigger + approved recovery route + wash-down receiving route where applicable + repeat-accumulation feedback	Ongoing performance feedback
VER-014	Wider applicability / rollout	A	G4	Demonstrated pilot/local result + location-specific mechanism/geometry/access review	Before standardization/rollout

Appendix E — Conceptual Engineering Drawings

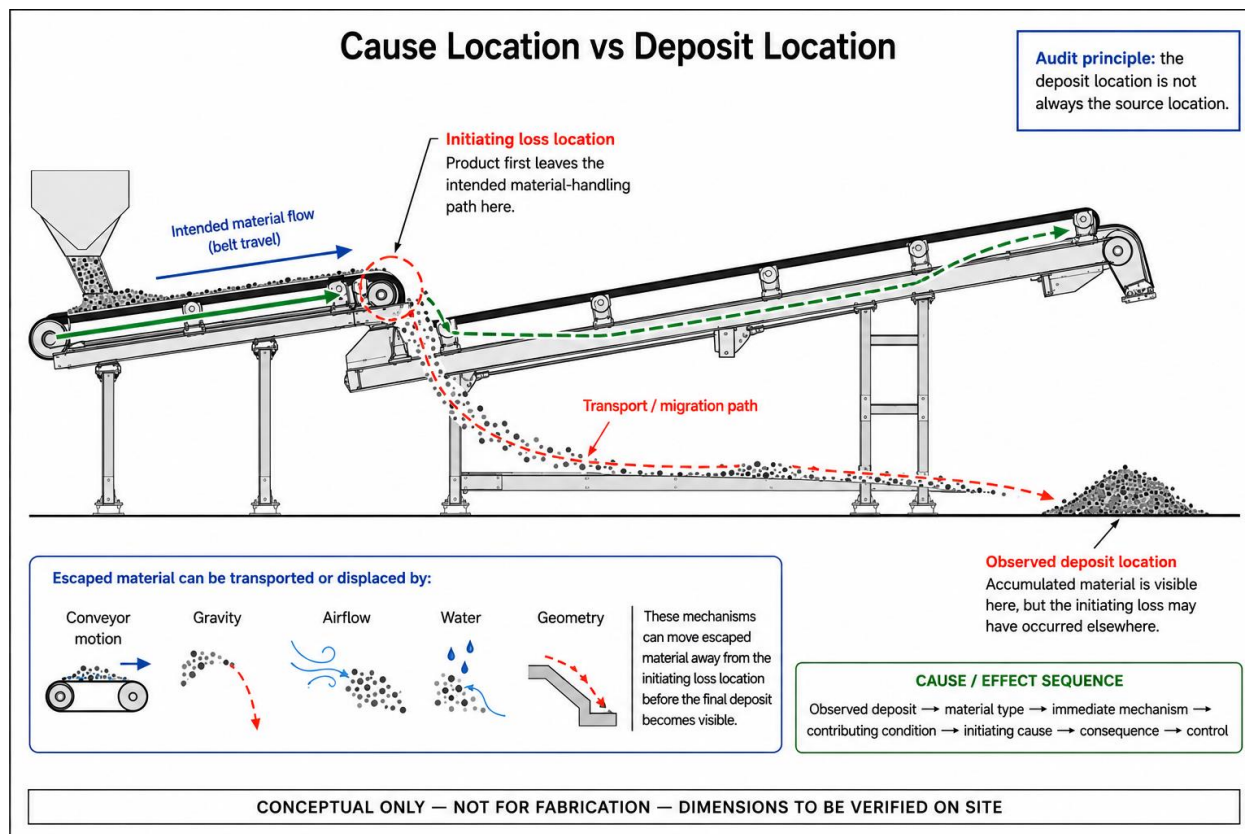
Appendix E controls and reproduces the current conceptual engineering drawings used in the report body. The six active report drawings are listed in Table 16 and reproduced below as reference copies so that the report remains technically self-contained. These appendix reproductions do not receive additional Figure numbers; Body Figures 1–6 remain the controlling report cross-references. Conceptual figures are report-support material, not as-found inspection evidence and not fabrication drawings. Historical CV09 CTD-001 and W10 CTD-002 remain listed for development traceability only and are not reproduced because they are superseded for active report use.

Table 16 — Conceptual engineering drawing register

ID	Title	Report use	Current control status
CTD-001	Historical CV09 Cleaner and Process-Containment Concept — SUPERSEDED	Historical traceability only	Superseded for active report use by CTD-005 and CTD-006.
CTD-002	Historical W10 Side-Containment Barrier Principle — SUPERSEDED	Historical traceability only	Superseded for active report use by CTD-010 and CTD-011; external walkway-barrier interpretation rejected.
CTD-003	Sample Conveyor Containment-Control Hierarchy	Body Figure 6 / Part V / Appendix E	Retained explanatory hierarchy only; not a final physical Sample Conveyor design.
CTD-004	Cause Location vs Deposit Location	Body Figure 1 / Part II / Appendix E	Retained explanatory audit principle; conceptual, not inspection evidence.
CTD-005	CV09 Preferred Primary / Secondary Cleaner Placement and Orientation	Body Figure 2 / Part IV / Appendix E	Accepted external/reference placement illustration retained for internal report use. Source: Dust Control Handbook for Industrial Minerals Mining and Processing, RI 9689 (2012). Source attribution retained; not presented as an ISS-original drawing.
CTD-006	CV09 Cleaners Inside Head-Chute Containment Principle	Body Figure 3 / Part IV / Appendix E	Accepted project conceptual figure: cleaners inside discharge chute so removed fines remain in the process flow.
CTD-010	W10 Side-Skirting / Product-Containment Concept	Body Figure 4 / Part V / Appendix E	Accepted W10 site-context/skirting-position concept; exact OEM detail remains open.
CTD-011	Conveyor Side Skirting / Spill Seal Concept — Cross-Section	Body Figure 5 / Part V / Appendix E	Accepted cross-sectional mechanism showing flexible containment inside carrying width; exact OEM detail remains open.

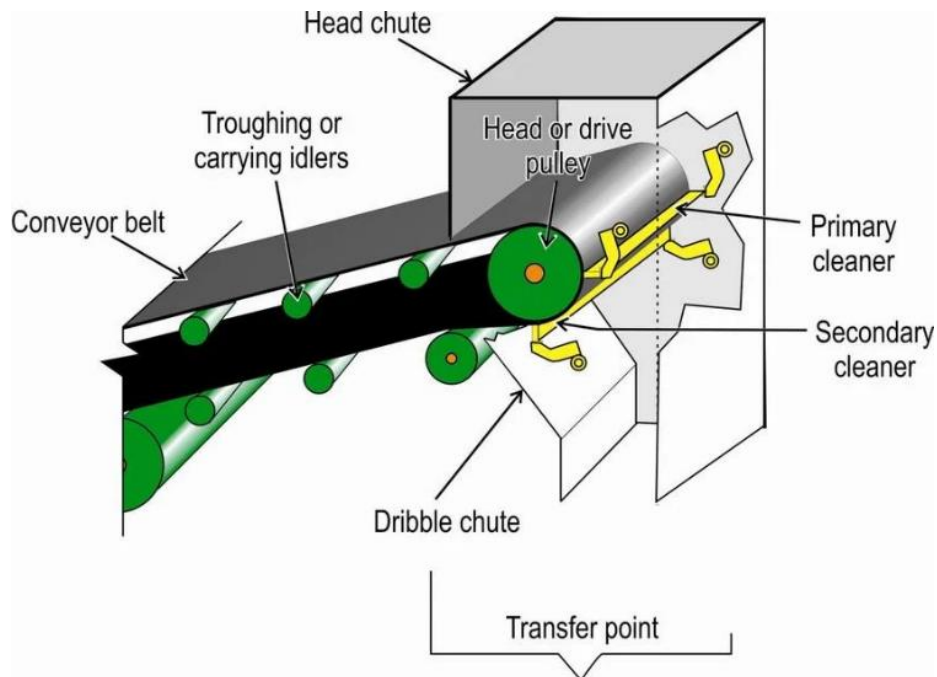
Active report drawings — appendix reference reproductions

Body Figure 1 / CTD-004 — Cause Location vs Deposit Location



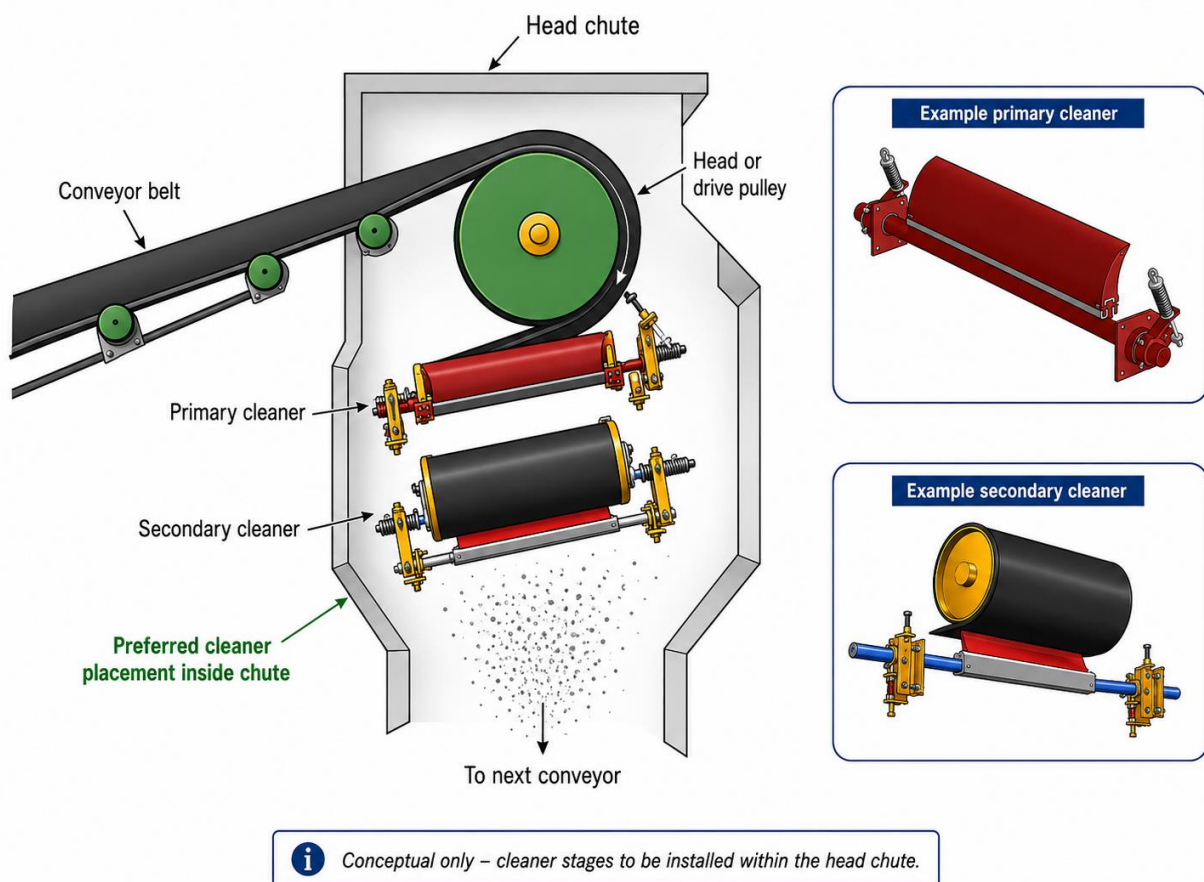
Status / use: Active explanatory audit-principle drawing. Conceptual only; not inspection evidence; not for fabrication.

Body Figure 2 / CTD-005 / DRAW-005 — CV09 Preferred Primary / Secondary Cleaner Placement and Orientation



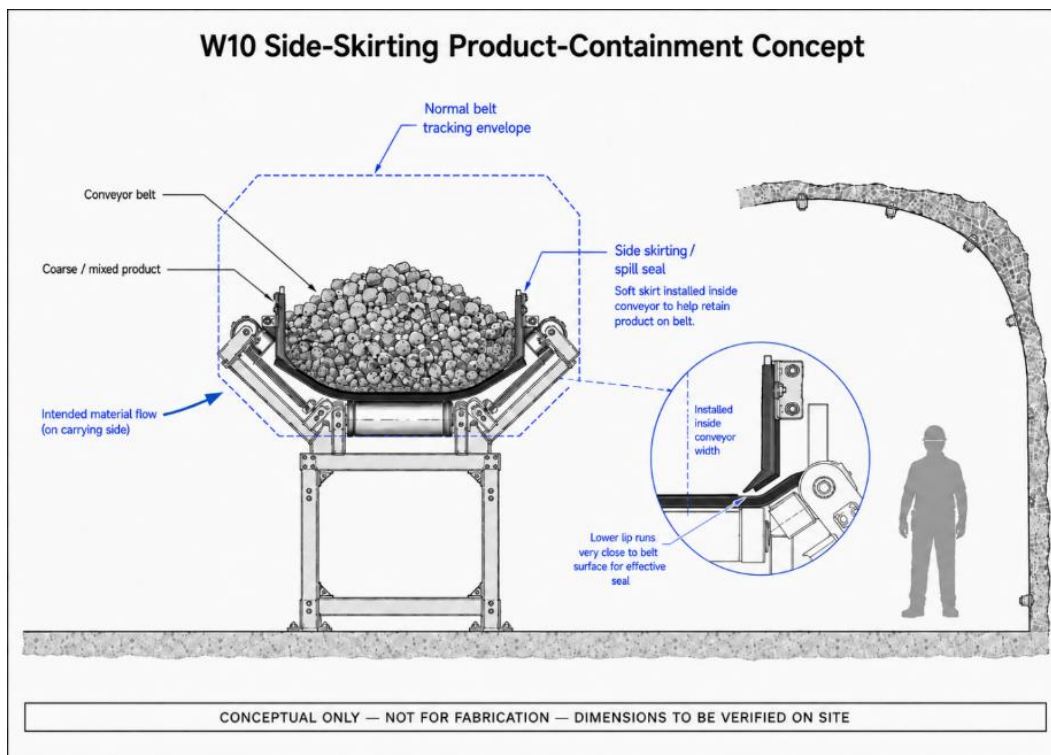
Status / use: Active external/reference placement illustration retained for internal report use. Source: Dust Control Handbook for Industrial Minerals Mining and Processing, RI 9689 (2012). Source attribution retained; not presented as an ISS-original drawing.

Body Figure 3 / CTD-006 / DRAW-006 — CV09 Cleaners Inside Head-Chute Containment Principle



Status / use: Active project conceptual drawing showing the cleaner stages inside the discharge-chute containment so removed fines remain within the process flow.

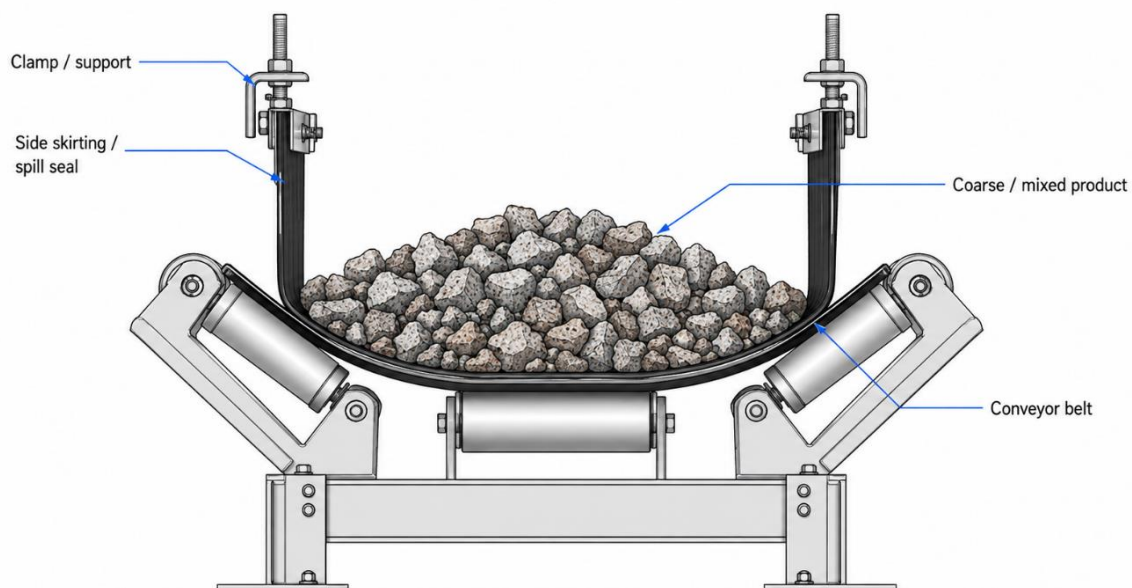
Body Figure 4 / CTD-010 / DRAW-0010 — W10 Side-Skirting / Product-Containment Concept



Status / use: Active W10 site-context concept showing where the side-skirt/spill-seal principle acts relative to the conveyor carrying edge. Exact OEM detail remains subject to verified site geometry and detailed design.

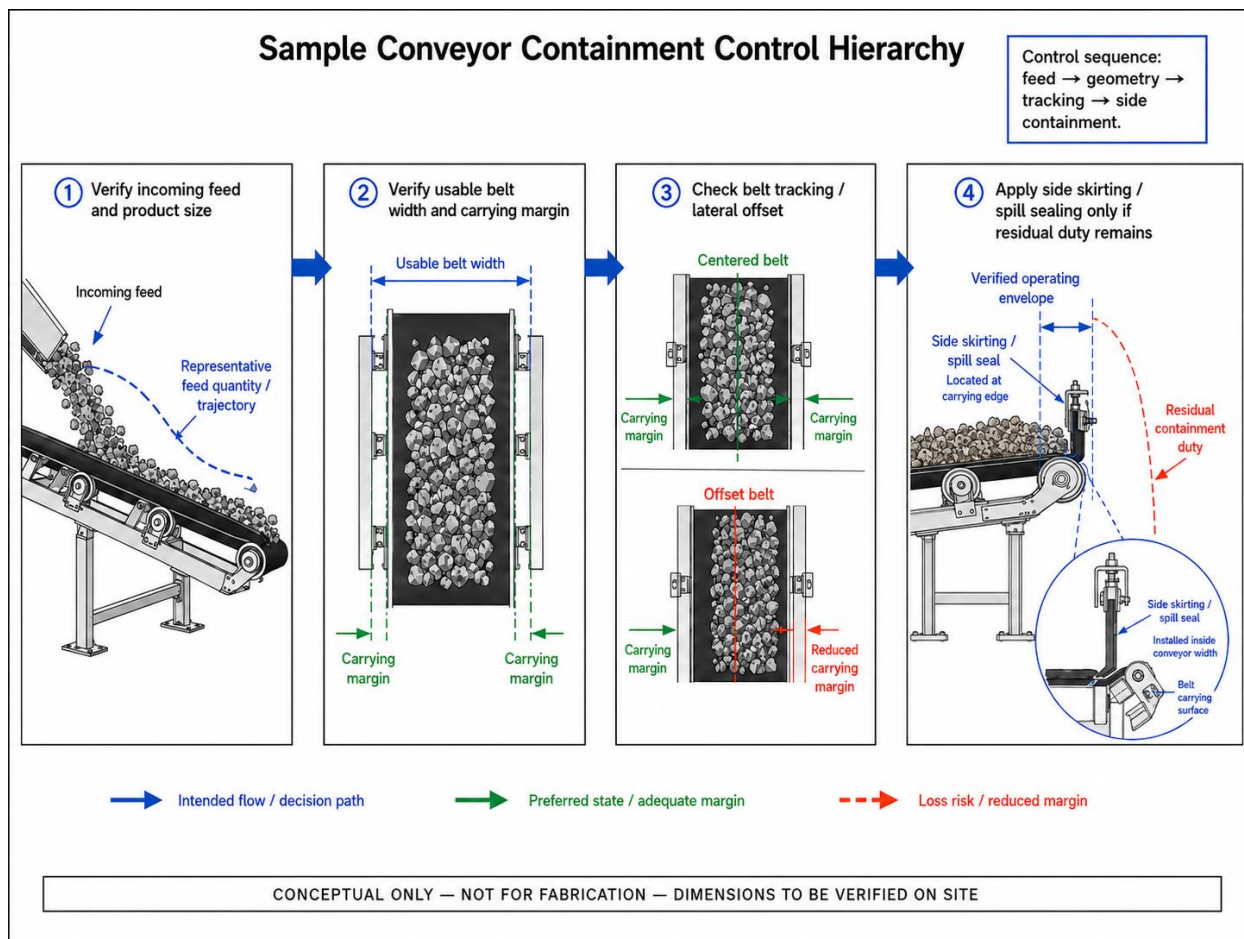
Body Figure 5 / CTD-011 / DRAW-0011 — Conveyor Side Skirting / Spill Seal Concept — Cross-Section

Conveyor Side Skirting / Spill Seal Concept (Cross-Section)



Status / use: Active cross-sectional mechanism showing flexible lateral containment inside the carrying width. Exact skirt material, contact arrangement and installation detail remain subject to OEM/responsible design.

Body Figure 6 / CTD-003 — Sample Conveyor Containment-Control Hierarchy



Status / use: Active explanatory hierarchy used in the Sample Conveyor discussion. Conceptual only; not for fabrication; site geometry and dimensions remain to be verified where applicable.

Appendix F — Source Records

Appendix F identifies the controlled source records used to develop, reconcile and finalize this report. Historical context and Phase 6/7 files remain preserved as the audit trail.

Context/Running_Report_Contextual_Spillage_NotesV4.3.md remains the consolidated contextual backbone; V4.3.1 governs the later system-level fines/CV09 functional-standard clarification; and V4.3.2 governs the current report-finalization and issue-control process. The Stage-A gap register, evidence_index_CURRENT.md and applicable active evidence overlays remain the controlled reconciliation and evidence-placement inputs.

Table 17 — Controlled source records register

Source class	Controlled record	Role / status
Inspection questionnaires	Context/Questionnaires/Running_Report_Contextual_Spillage_Questionnaire_ResponseV1.docx; V2.docx	Source-recorded inspection/operational context; V2 is later expanded source.
Current controlled context chain	Context/Running_Report_Contextual_Spillage_NotesV4.3.md; V4.3.1_System_Fines_Containment_Alignment_Addendum.md; V4.3.2_Finalisation_Control_Addendum.md	Context V4.3 contextual backbone + V4.3.1/V4.3.2 controlled addenda + Stage-A Gap Register + current evidence controls.
Phase 6 historical addendum	Context/Running_Report_Contextual_Spillage_NotesV3.1_0_Phase6_Addendum.md	Historical Phase 6 assembly context; preserved, not overwritten.
Phase 7 addendum	Context/Running_Report_Contextual_Spillage_NotesV3.1_1_Phase7_Addendum.md	Phase 7 refinement/control context retained in the current context chain.
Current evidence index	Report-Working/evidence_index_CURRENT.md	Current controlled evidence population: 196 media / 188 photographs / 8 videos / 79 E / 40 S / 77 A, with placement governed by evidence_index_CURRENT.md, active overlays and the REF-013 withdrawal control.
Visual-review records	Report-Working/evidence_visual_review_batch_A.md through _F.md	Detailed per-item visual evidence strength, limitation and qualification.
Evidence/story matrix	Report-Working/SpillageReport/EVIDENCE_STORY_MATRIX_V0.1.csv / .md	Historical 197-item V0.1 base matrix retained as audit trail; later asset overlays and REF-013 withdrawal supersede applicable rows for current report use.
Narrative architecture / outline	REPORT_NARRATIVE_ARCHITECTURE.md; REPORT_NARRATIVE_OUTLINE_V0.1.md	Accepted report identity, Parts I-XII journey and mechanism-based narrative.
Phase 6 closure	PHASE6_ASSEMBLY_CLOSURE.md	Inspector approval of V0.2 assembly architecture.
Phase 7 content set	PHASE7_FRONT_END_REFINEMENT_DRAFT_V0.1.md; Part IV-VIII drafts; Part IX; Part X; Part XI V0.3; Part XII	Report-ready narrative/action content carried into V0.5; technical substance retained during language and terminology harmonization.
Verification register	PHASE7_VERIFICATION_SUCCESS_CRITERIA_REGISTER_V0.1.md	Controlled VER-001..VER-014 framework; current report wording aligned to the system-level fines-containment standard, progressive applicability programme and current governance/verification boundaries.
Current concept figure register	Report-Working/SpillageReport/PHASE7_CONCEPTUAL_ENGINEERING_FIGURE_REGISTER_V0.3.md	Current figure control: CTD-001/002 superseded; CTD-005/006 active CV09; CTD-010/011 active W10; CTD-003/004 retained explanatory concepts.
Phase 7 consistency / closure	PHASE7_PASS7F_TECHNICAL_CONSISTENCY_REVIEW_V0.1.md; PHASE7_REFINEMENT_CLOSURE.md	Final evidence-to-action consistency PASS and Phase 7 content closure.
Language-style control addendum	Context/Running_Report_Contextual_Spillage_NotesV3.1_2_Language_Style_Addendum.md	Current binding reader-facing language rule; V0.2 softer formal-engineering narrative is the tonal benchmark.
Terminology control addendum	Context/Running_Report_Contextual_Spillage_NotesV3.1_3_Terminology_Refinement_Addendum.md	Current binding terminology rule: use direct physical path/process language in reader-facing narrative; retain "envelope" only for defined geometric/operating ranges.
Report lineage	V0.1 working base; V0.2 Assembly Draft; V0.3 Compilation Draft; V0.4 Language-Harmonized Draft; V0.5 Terminology-Refined Draft; V0.6 Stage-B Integrated Controlled Draft; V0.6.1 System-Level Fines-Containment Alignment Draft; V1.x finalization sequence culminating in Revision 1.5 controlled issue	Context V4.3 contextual backbone + V4.3.1/V4.3.2 controlled addenda + Stage-A Gap Register + current evidence controls. Finalization baseline; final document-control/revision synchronization, field/navigation refresh, publication QA and formal Report lineage closed at Revision 1.5 controlled issue.
Stage-A reconciliation control	Context/Running_Report_Contextual_Spillage_NotesV4.1_Report_Gap_Register_StageA.md	Diagnostic V0.5 ↔ source-reconciled-context gap register retained as a controlled V0.6/V0.6.1 integration input.
Current active evidence overlays	EVIDENCE_STORY_MATRIX_V0.2_SAMPLE_CONVEYORS.csv; V0.3_U4_L3.csv; V0.4_SCREENERS.csv; V0.5_L1_STRUCTURAL.csv; SAMPLE_CONVEYOR3_REF013_EVIDENCE_WITHDRAWAL.md	Supersede applicable earlier asset rows and retain current body/story priorities for the issued report.
Indicative supplier / market reference	Tru-Trac Rollers (Pty) Ltd — Belt Scrapers / Conveyor Belt Cleaners — tru-trac.com	Indicative supplier reference only; supports supplier availability/capability context for primary and secondary belt-cleaner systems. Not inspection evidence, not a prescribed product/model and not a sole-source procurement requirement.

Appendix G - Complete Inspection Photographic Evidence Catalogue

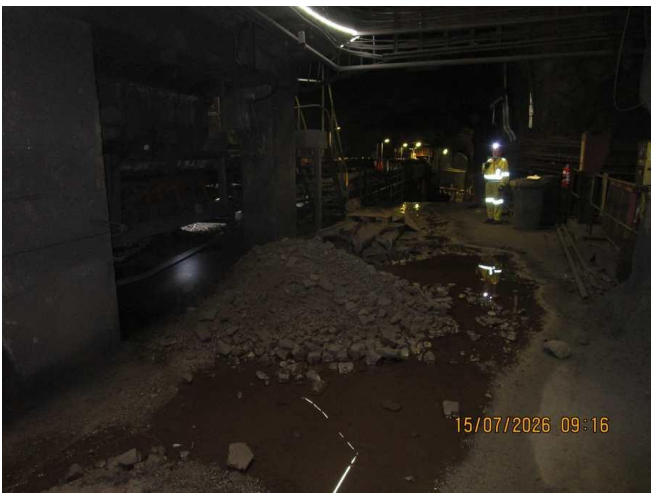
S32WPPSR-01 | South32 Wessels Mine Combined-Product Spillage Audit

This appendix presents reduced-resolution thumbnails of the 188 active inspection photographs contained in the controlled evidence population. It is provided as a compact visual catalogue to support evidence traceability and orientation. Original/high-resolution photographs remain in the controlled evidence repository and selected higher-quality evidence is reproduced in the report body where required. Inspection videos and conceptual/technical illustrations are excluded from this catalogue. Withdrawn/non-active evidence is not reproduced. Canonical evidence IDs shown below correspond with the controlled photographic evidence register in Appendix B.

188 Active photographs	2 columns Compact visual grid	Reduced resolution Size-controlled appendix
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UNDERGROUND INSPECTION PHOTOGRAPHS

Underground Crusher | 5 photographs



UGCrusher-REF-001



UGCrusher-REF-002



UGCrusher-REF-003



UGCrusher-REF-004

Underground Crusher | 5 photographs



UGCrusher-REF-005

UC05 / UC04 / UC03 / UC02 / UC01 continuous conveyor system | 16 photographs



UGUC05_04_03_02_01-REF-001



UGUC05_04_03_02_01-REF-002

UC05 / UC04 / UC03 / UC02 / UC01 continuous conveyor system | 16 photographs



UGUC05_04_03_02_01-REF-003



UGUC05_04_03_02_01-REF-004



UGUC05_04_03_02_01-REF-005



UGUC05_04_03_02_01-REF-006

UC05 / UC04 / UC03 / UC02 / UC01 continuous conveyor system | 16 photographs



UGUC05_04_03_02_01-REF-007



UGUC05_04_03_02_01-REF-008



UGUC05_04_03_02_01-REF-009



UGUC05_04_03_02_01-REF-010

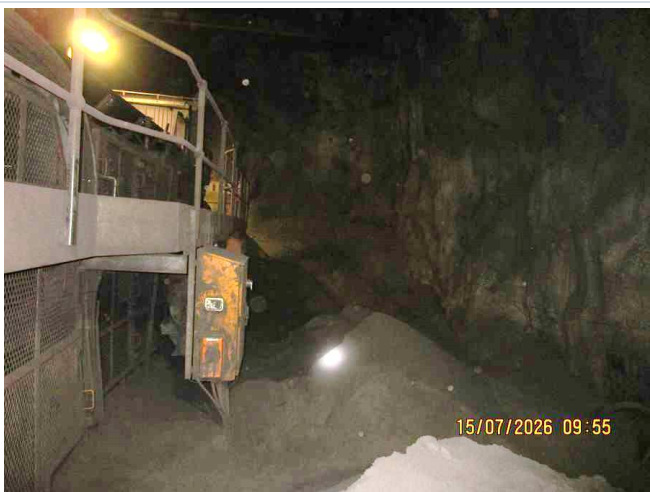
UC05 / UC04 / UC03 / UC02 / UC01 continuous conveyor system | 16 photographs



UGUC05_04_03_02_01-REF-011



UGUC05_04_03_02_01-REF-012



UGUC05_04_03_02_01-REF-013



UGUC05_04_03_02_01-REF-014

UC05 / UC04 / UC03 / UC02 / UC01 continuous conveyor system | 16 photographs



UGUC05_04_03_02_01-REF-015



UGUC05_04_03_02_01-REF-016

U3 Conveyor | 8 photographs



UGU3-REF-001



UGU3-REF-002

U3 Conveyor | 8 photographs



UGU3-REF-003



UGU3-REF-004



UGU3-REF-005



UGU3-REF-006

U3 Conveyor | 8 photographs



UGU3-REF-007



UGU3-REF-008

U4 Tripper Car | 11 photographs



UGU4-REF-001



UGU4-REF-002

U4 Tripper Car | 11 photographs



UGU4-REF-003



UGU4-REF-004



UGU4-REF-005

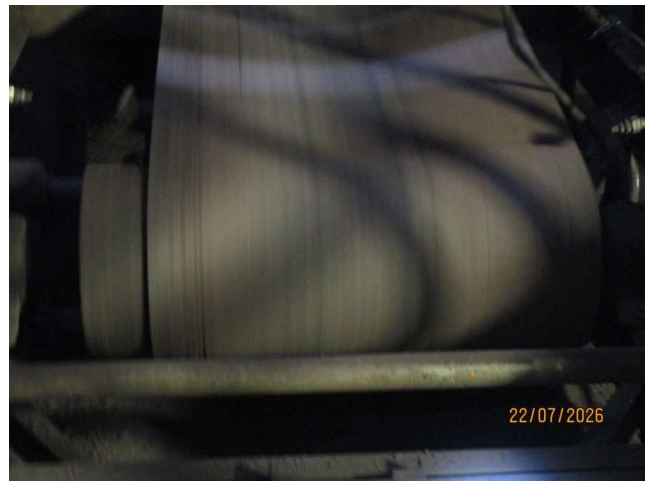


UGU4-REF-006

U4 Tripper Car | 11 photographs



UGU4-REF-007



UGU4-REF-008



UGU4-REF-009



UGU4-REF-011



UGU4-REF-012

U5A Conveyor | 9 photographs



UG5A-REF-001



UG5A-REF-002

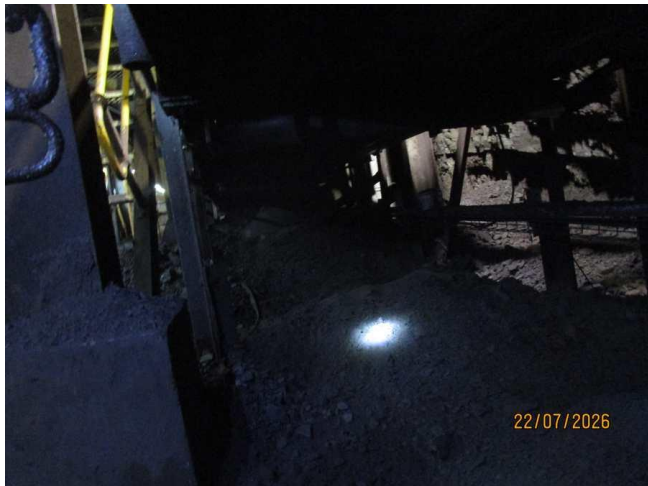


UG5A-REF-003



UG5A-REF-004

U5A Conveyor | 9 photographs



UG5A-REF-005



UG5A-REF-006



UG5A-REF-007



UG5A-REF-008

U5A Conveyor | 9 photographs



UG5A-REF-009

U5B Conveyor | 6 photographs



UG5B-REF-001



UG5B-REF-002

U5B Conveyor | 6 photographs



UG5B-REF-003



UG5B-REF-004



UG5B-REF-005



UG5B-REF-006

SURFACE INSPECTION PHOTOGRAPHS

W10 Conveyor | 12 photographs



W10-REF-001



W10-REF-002

W10 Conveyor | 12 photographs



W10-REF-003



W10-REF-004

W10 Conveyor | 12 photographs



W10-REF-005



W10-REF-006



W10-REF-007



W10-REF-008

W10 Conveyor | 12 photographs



W10-REF-009



W10-REF-010



W10-REF-011



W10-REF-012

CV01 Conveyor | 4 photographs



CV01-REF-003



CV01-REF-002

CV01 Conveyor | 4 photographs



CV01-REF-001



CV01-REF-004

CV02 Conveyor | 4 photographs



CV02-REF-001



CV02-REF-002

CV02 Conveyor | 4 photographs



CV02-REF-003



CV02-REF-004

CV03 Conveyor | 3 photographs



CV03-REF-001



CV03-REF-002

CV03 Conveyor | 3 photographs



CV03-REF-003

CV04 Conveyor | 5 photographs



CV04-REF-002



CV04-REF-003



CV04-REF-004

CV04 Conveyor | 5 photographs



CV04-REF-005

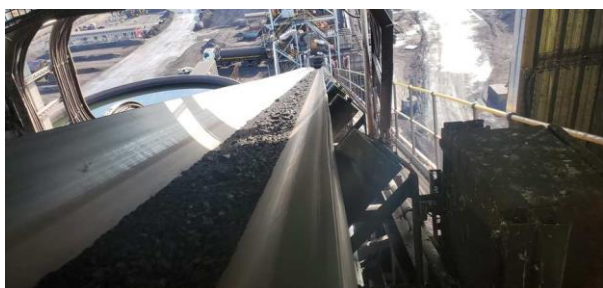
CV06 Conveyor | 5 photographs



CV06-REF-001



CV06-REF-002



CV06-REF-003



CV06-REF-004

CV06 Conveyor | 5 photographs



CV06-REF-005

CV09 Conveyor | 5 photographs



CV09-REF-001



CV09-REF-004

CV09 Conveyor | 5 photographs



CV09-REF-002



CV09-REF-003

CV09 Conveyor | 5 photographs



CV09-REF-005

CV14 Conveyor | 2 photographs



CV14-REF-001



CV14-REF-002

CV15 Conveyor | 8 photographs



CV15-REF-001



CV15-REF-002

CV15 Conveyor | 8 photographs



CV15-REF-005



CV15-REF-004



CV15-REF-006



CV15-REF-008

Screens / Screen Chutes | 14 photographs



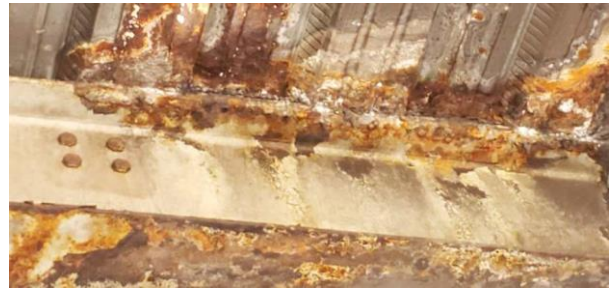
Screen-REF-001



Screen-REF-002



Screen-REF-003



Screen-REF-004



Screen-REF-005



Screen-REF-006



Screen-REF-007



Screen-REF-008



Screen-REF-009



Screen-REF-010

Screens / Screen Chutes | 14 photographs**Screen-REF-011****Screen-REF-012****Screen-REF-013****Screen-REF-014****Loading Station - L1 Conveyor | 18 photographs****LoadoutStationL1-REF-001****LoadoutStationL1-REF-002****LoadoutStationL1-REF-003****LoadoutStationL1-REF-004****LoadoutStationL1-REF-005****LoadoutStationL1-REF-006**

Loading Station - L1 Conveyor | 18 photographs



LoadoutStationL1-REF-007



LoadoutStationL1-REF-008



LoadoutStationL1-REF-009



LoadoutStationL1-REF-010



LoadoutStationL1-REF-011



LoadoutStationL1-REF-012



LoadoutStationL1-REF-013



LoadoutStationL1-REF-014



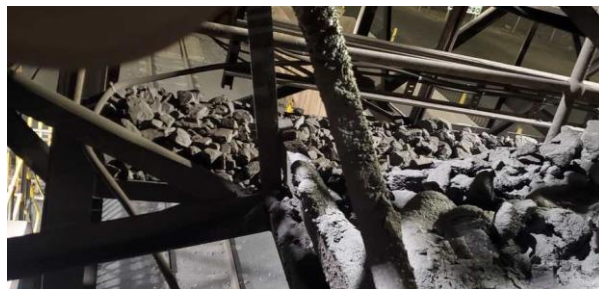
LoadoutStationL1-REF-015



LoadoutStationL1-REF-016

Loading Station - L1 Conveyor | 18 photographs

LoadoutStationL1-REF-017



LoadoutStationL1-REF-018

Loading Station - L2 Conveyor | 6 photographs

LoadoutStationL2-REF-001



LoadoutStationL2-REF-002



LoadoutStationL2-REF-003



LoadoutStationL2-REF-004



LoadoutStationL2-REF-005



LoadoutStationL2-REF-006

Loading Station - L3 Conveyor | 8 photographs

LoadoutStationL3-REF-001

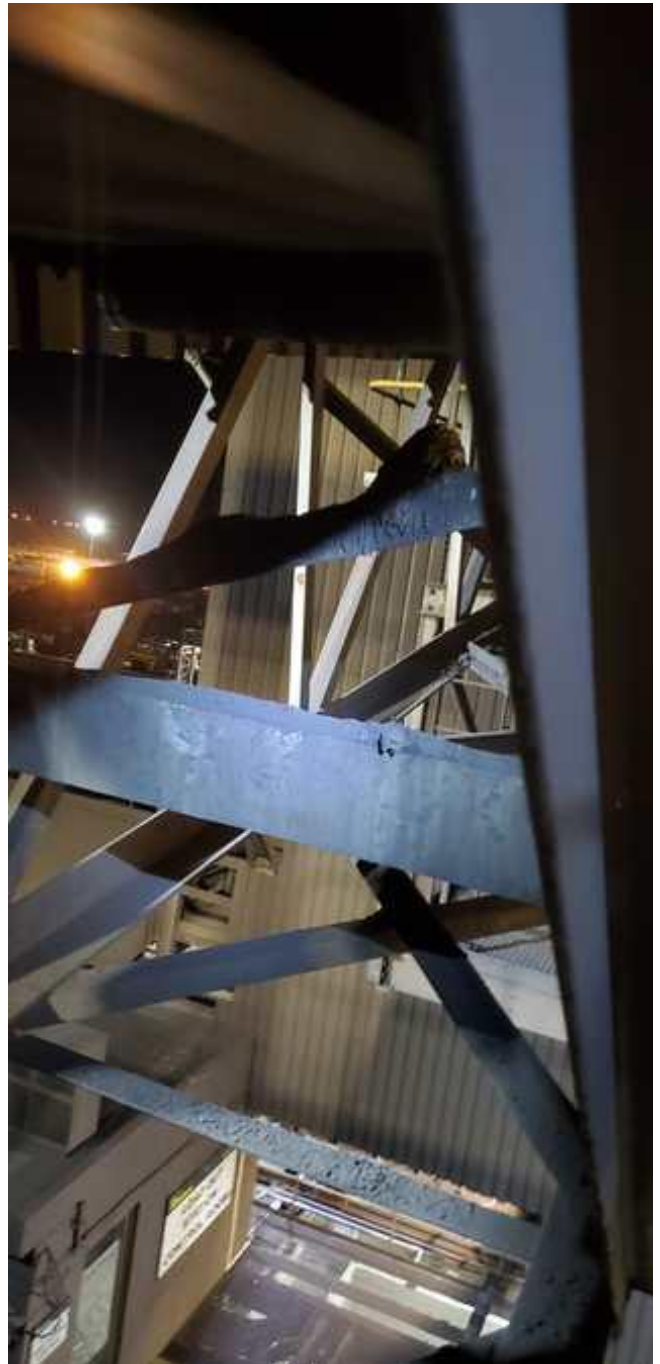


LoadoutStationL3-REF-002

Loading Station - L3 Conveyor | 8 photographs



LoadoutStationL3-REF-003



LoadoutStationL3-REF-004

Loading Station - L3 Conveyor | 8 photographs



LoadoutStationL3-REF-005



LoadoutStationL3-REF-006

Loading Station - L3 Conveyor | 8 photographs



LoadoutStationL3-REF-007

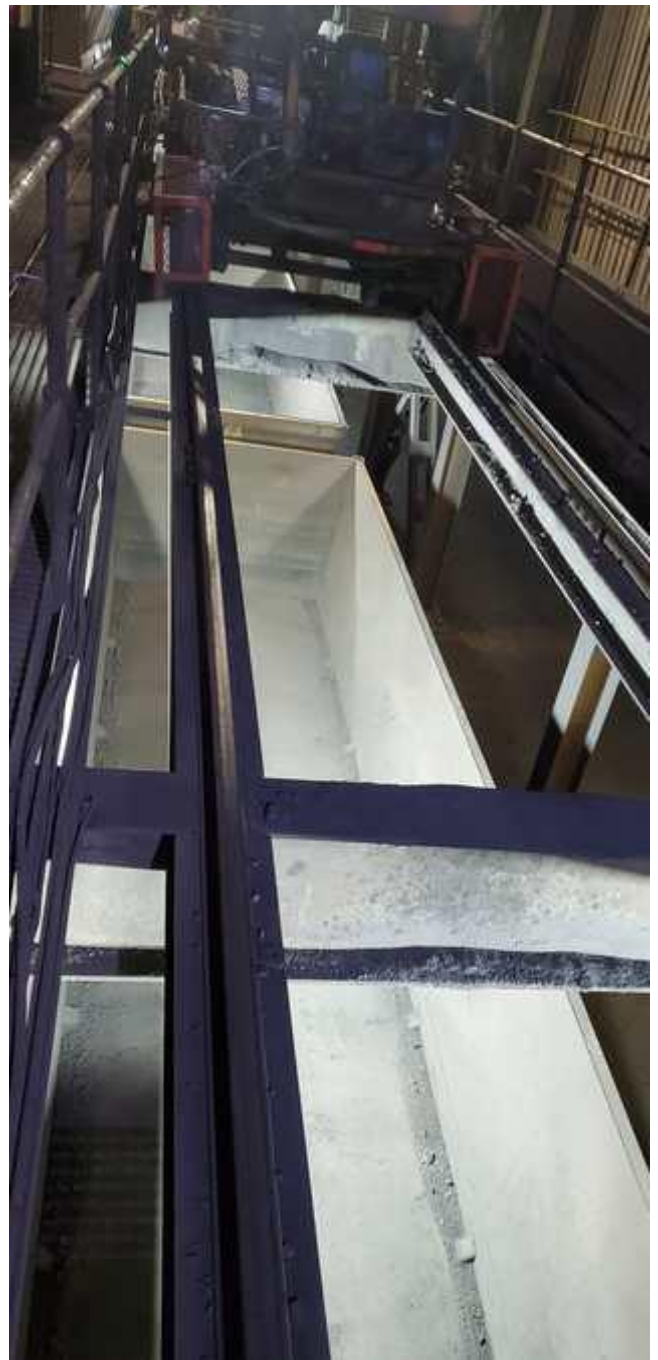


LoadoutStationL3-REF-008

Loading Station - L4 Conveyor | 8 photographs



LoadoutStationL4-REF-001



LoadoutStationL4-REF-002

Loading Station - L4 Conveyor | 8 photographs



LoadoutStationL4-REF-003



LoadoutStationL4-REF-004

Loading Station - L4 Conveyor | 8 photographs



LoadoutStationL4-REF-005



LoadoutStationL4-REF-006



LoadoutStationL4-REF-007



LoadoutStationL4-REF-008

Loading Station - Sample Conveyor 1 | 11 photographs



LoadoutStationSample1-REF-001



LoadoutStationSample1-REF-002



LoadoutStationSample1-REF-003



LoadoutStationSample1-REF-004



LoadoutStationSample1-REF-005

Loading Station - Sample Conveyor 1 | 11 photographs



LoadoutStationSample1-REF-006

Loading Station - Sample Conveyor 1 | 11 photographs



LoadoutStationSample1-REF-007



LoadoutStationSample1-REF-008



Loading Station - Sample Conveyor 1 | 11 photographs

LoadoutStationSample1-REF-010



LoadoutStationSample1-REF-011



LoadoutStationSample1-REF-009

Loading Station - Sample Conveyor 2 | 8 photographs



LoadoutStationSample2-REF-001



LoadoutStationSample2-REF-002

Loading Station - Sample Conveyor 2 | 8 photographs



LoadoutStationSample2-REF-003



LoadoutStationSample2-REF-004



LoadoutStationSample2-REF-005



LoadoutStationSample2-REF-006

Loading Station - Sample Conveyor 2 | 8 photographs



LoadoutStationSample2-REF-007



LoadoutStationSample2-REF-008

Loading Station - Sample Conveyor 3 | 12 photographs



LoadoutStationSample3-REF-001



LoadoutStationSample3-REF-002

Loading Station - Sample Conveyor 3 | 12 photographs



LoadoutStationSample3-REF-003

Loading Station - Sample Conveyor 3 | 12 photographs



LoadoutStationSample3-REF-004

Loading Station - Sample Conveyor 3 | 12 photographs



LoadoutStationSample3-REF-005

Loading Station - Sample Conveyor 3 | 12 photographs



LoadoutStationSample3-REF-006



LoadoutStationSample3-REF-007

Loading Station - Sample Conveyor 3 | 12 photographs



LoadoutStationSample3-REF-008



LoadoutStationSample3-REF-009



LoadoutStationSample3-REF-010

Loading Station - Sample Conveyor 3 | 12 photographs



LoadoutStationSample3-REF-011



LoadoutStationSample3-REF-012

Evidence-control note: This visual catalogue is supplementary to the controlled evidence registers. Where a thumbnail is insufficient for detailed assessment, the original evidence file should be consulted.